Regional Workshop on Updating and Advancing the Initial Pipeline of ASEAN Infrastructure Projects

9 – 10 April 2025

9th April 2025
Workshop – Day 1

Agenda

Time (GMT+8)	Agenda	Format
0830 – 0900	Registration for Day 1	
0900 – 0905	Opening remarks LIB-SI Chair	Plenary
0905 – 0910	Opening remarks by representative of the Government of Australia	Plenary
0910 – 0915	Photo Session	Plenary
0915 – 1000	SMBC presentation and Q&A on the Project overview, status, challenges, and interim findings	Plenary
1000 – 1030	Informational session: SMBC presentation and Q&A on the state of the financing market (ECAs, DFIs, commercial lenders) and general bankability considerations	Plenary
1030 – 1045	Coffee break	
1045 – 1155	Presentations by project owners on potential projects (3-4 projects) (Note: 15min presentation + 5min Q&A for each Project)	Plenary
1155 – 1330	Lunch break	

Agenda

Time (GMT+8)	Agenda	Format
1330 – 1400	Informational session: SMBC presentation and Q&A on key developments, challenges, opportunities in transportation infrastructure in the region, and general bankability considerations (such as with examples of relevant connectivity projects financed in recent years)	Plenary
1400 – 1515	Brainstorming and deep dive on the projects presented, including feedback and suggestions on improving the projects	Breakout
1515 – 1545	Discussion of feedbacks and suggestions from Breakout session	Plenary
1545 – 1600	Coffee break	
1600 – 1630	Informational session: SMBC presentation and Q&A on key developments, challenges, and opportunities in digital infrastructure in the region	Plenary
1630 – 1640	Wrapping up of Day 1	Plenary

Opening Remarks by

H.E. Dato’ Nor Azmie Bin Diron
LIBSI Chair
Secretary General, Ministry of Economy, Malaysia

Opening Remarks by

H.E. Tiffany McDonald

Australian Ambassador To ASEAN

Photo Session

Project overview, status, challenges, and interim findings

SMBC Support to ASEAN

As a recap, SMBC was appointed as consultant for the Project, “Advancing and Updating the Initial Rolling Priority Pipeline of ASEAN Infrastructure Projects through Innovative Financial Approaches” under the ambit of LIB-SI, through a tender process.

The primary objectives are:

- Advancing and updating the Initial Pipeline of ASEAN infrastructure projects
- Identifying appropriate financing approaches for selected projects
- Garnering market interest in the selected projects

Project Timeline

Inception Report for the Project has been endorsed by LIB-SI in November 2023.

SMBC is looking to conclude the information gathering and screening of project submissions by April 2025.

Information Gathering	Country Consultation	Screening Process	Regional Workshop	Financing Approach	Market Sounding / Project Promotion	
Jan 2024 – Apr 2025	Feb 2024 – Apr 2025	Apr 2024 – Apr 2025	Apr 2025	Apr – May 2025	May – Jun 2025	Jun 2025
Project Submissions to be completed by Apr 2024			**Draft Interim Report by Apr 2025**	**Project Brief on the 8 shortlisted projects is expected to be ready by May 2025**	**Market Sounding Report by Jun 2025**	**Final Report and Project Completion Report by Jun 2025 (Extension to be discussed)**

Information Gathering Status

Summary of our activities

Throughout the Information Gathering process, the SMBC Team has:

- Conducted **27** country consultations
- Made **7** trips for in-country consultations
- Spoken to a total of **~180** personnel

Key Results

- We have received a total of **18** Project Application Forms to date, of which:
  - **13** projects in the roads sector
  - **3** projects in the digital infrastructure
  - **1** project in airport development
  - **1** project in education sector
- For the Initial Pipeline Projects, we have also gathered project updates on **10 out of the 19 projects** and is following up on the remaining projects

We note that some AMS require more time for preparation of submission, and we have thus extended the submission deadline to the end of April 2025.

Updates to Initial Pipeline Project (1/2)

The Shortlisted projects will be added to the Initial Pipeline Project to form the Updated ASEAN Infrastructure Pipeline. We have gathered feedback on updates to the Initial Pipeline Projects for a few projects as per below and we would request for AMS LIB-SI to provide updates on the remainder of the projects.

No.	Project Name	Country	Status [To be Updated]
1	Jalan Rasau Road Upgrading	Brunei	LIB-SI Brunei has requested to drop this project from the pipeline.
2	Siem Reap to Ratanakiri Road Upgrading	Cambodia	LIB-SI Cambodia has updated that the project is at the stage of the Feasibility Study (FS) and Detail Engineering Design (DED). Financing for the project is expected to be approved by end of 2026.
3	Kuala Tanjung International Hub Port and Industrial Estates - Phase II	Indonesia	Project is now operating and is working to improve the use of its Liquid Bulk Terminal
4	Expansion of Hang Nadim International Airport	Indonesia	Project has been successfully implemented and would be dropped off from the pipeline
5	Development of Kijing Port	Indonesia	Project is under progress
6	Lao PDR National Road No. 2 Upgrading	Lao PDR	Project has secured funding
7	Lao PDR National Road No. 8 Upgrading	Lao PDR	Project is seeking funding
8	Lao PDR - Viet Nam Power Interconnector	Lao PDR	LIB-SI Lao PDR is waiting for updates from the relevant ministry
9	Lao PDR - Myanmar Power Interconnector (Lao PDR section)	Lao PDR	LIB-SI Lao PDR is waiting for updates from the relevant ministry

Updates to Initial Pipeline Project (2/2)

No.	Project Name	Country	Status [To be Updated]
10	Myanmar - Lao PDR Power Interconnector (Myanmar section)	Myanmar	Pending further updates from LIB-SI Myanmar.
11	Yangon - Mandalay Expressway	Myanmar	LIB-SI Myanmar has updated that this remains a priority project.
12	Nay Pyi Taw - Kyaukpyu Expressway	Myanmar	LIB-SI Myanmar has requested to drop this project from the pipeline.
13	Muse - Tigyaing - Mandalay Expressway	Myanmar	LIB-SI Myanmar has requested to drop this project from the pipeline.
14	Tarlay - Kyainglat Road Upgrading	Myanmar	LIB-SI Myanmar has requested to drop this project from the pipeline.
15	ASEAN Digital Hub	Thailand	LIB-SI Thailand updated that this project should be completed by 2025.
16	Hat Yai - Sadao Motorway	Thailand	LIB-SI Thailand updated that this project is currently undergoing a feasibility study. This project has been submitted for consideration in the Updated Pipeline.
17	Bangkok - Nong Khai HSR - Phase 2	Thailand	LIB-SI Thailand updated that this project has been approved by the Cabinet on 4th February.
18	Southern Coastal Corridor Project - Phase II	Vietnam	LIB-SI Vietnam updated that the project is currently in the process of finalizing the investment decision, selecting contractors through a bidding process, and implementing land clearance.
19	Ho Chi Minh City - Moc Bai Expressway	Vietnam	LIB-SI Vietnam has requested to drop this project from the pipeline.

Project Screening and Shortlisting

Modifications have been implemented to the shortlisting criteria, which follows a two-stage process in screening and shortlisting of the projects.

Activity	Criteria	Work product
Stage 1 – project screening	▶ Connectivity infrastructure ▶ Pre-procurement stage of development ▶ Regional impact ▶ Identified as a priority by at least one AMS ▶ Sufficient information	Is a pass/fail evaluation. Only projects that pass the initial screening will be evaluated further in detail for inclusion in the list of 20 projects in the updated pipeline
Stage 2 – project selection	▶ Strategic Relevance ▶ Regional Connectivity ▶ Resilience ▶ Economic Environmental & Social Benefits ▶ Project Feasibility ▶ Implementation Capacity	Each project that clears the initial screening will be evaluated in detail and the top 20 projects will be evaluated further for inclusion in the Updated Pipeline.

Project Screening and Shortlisting

The scoring criteria have been structured for greater emphasis on strategic relevance and economic, environmental, and social benefits as shown below:

ASEAN Infrastructure Rolling Pipeline Scoring		
Section		
Basic Information	Not scored	
Strategic Relevance	40	23%
Regional Connectivity	18	10%
Resilience	36	20%
Economic, Environmental & Social benefits	40	23%
Project Feasibility	30	17%
Implementation Capacity	12	7%
Total score:	**176**	**100%**

Proposed Updated Project Shortlist

Based on the screening criteria, we have proposed for the below projects to be shortlisted for addition into the ASEAN Infrastructure Pipeline. We are in the process of evaluating a few other projects with potential to be included in the Updated Pipeline as well.

No.	Country Name	Project Name	Project Description	Projected Cost (USD'mil)	Proposed project timeline	Evaluation comments
1	Philippines	Development of an ASEAN Space Data Infrastructure	Greenfield, early concept stage, digital infra	389	2026 – 2030	High impact on connectivity among ASEAN countries, brings a new dimension to connectivity
2	Thailand	Hatyai – Sadao Motorway	Greenfield, feasibility/ funding identified, transport	903	2028 - 2031	High impact on connectivity, well-progressed project with feasibility study and detailed design stage completed
3	Thailand	Highway connecting Udon Thani – Bueng Kan	Greenfield, funding identified, transport	577	2028 - 2031	High impact on connectivity, well-progressed project, support from multilateral being discussed
4	Thailand	Don Mueang Development Project Phase 3	Brownfield, funding identified, transport	1,116	2027 - 2033	High impact on connectivity among ASEAN countries, funding identified, government supporting land acquisition
5	Thailand	Landbridge Project	Greenfield, feasibility stage, Port	27,000	2026 - 2030	High impact on connectivity among ASEAN countries, highly interested by private sector
6	Myanmar	Upgradation of Meiktila – Taungyi- Keng Tung – Tachileik Road	Brownfield, early concept stage, transport	331	2025 – 2028	Government funded, no conflict zone, high impact

Combined Project Pipeline (1/3)

Below presents the combined list of projects from the Updated and Initial Pipeline.

No.	Country Name	Project Name	Initial Pipeline/ Updated Pipeline
1	Philippines	Development of an ASEAN Space Data Infrastructure	Updated Pipeline
2	Thailand	Hatyai – Sadao Motorway	Updated Pipeline
3	Thailand	Highway connecting Udon Thani – Bueng Kan	Updated Pipeline
4	Thailand	Don Mueang Development Project Phase 3	Updated Pipeline
5	Thailand	Landbridge Project	Updated Pipeline
6	Myanmar	Upgradation of Meiktila – Taungyi- Keng Tung – Tachileik Road	Updated Pipeline

Combined Project Pipeline (2/3)

Below presents the combined list of projects from the Updated and Initial Pipeline.

No.	Country Name	Project Name	Initial Pipeline/ Updated Pipeline
7	Brunei	Jalan Rasau Road Upgrading	Initial Pipeline
8	Cambodia	Siem Reap to Ratanakiri Road Upgrading	Initial Pipeline
9	Indonesia	Kuala Tanjung International Hub Port and Industrial Estates - Phase II	Initial Pipeline
10	Indonesia	Development of Kijing Port	Initial Pipeline
11	Lao PDR	Lao PDR National Road No. 2 Upgrading	Initial Pipeline
12	Lao PDR	Lao PDR National Road No. 8 Upgrading	Initial Pipeline
13	Lao PDR	Lao PDR - Viet Nam Power Interconnector	Initial Pipeline
14	Lao PDR	Lao PDR - Myanmar Power Interconnector (Lao PDR section)	Initial Pipeline

Combined Project Pipeline (3/3)

Below presents the combined list of projects from the Updated and Initial Pipeline.

No.	Country Name	Project Name	Initial Pipeline/ Updated Pipeline
15	Myanmar	Myanmar - Lao PDR Power Interconnector (Myanmar section)	Initial Pipeline
16	Myanmar	Yangon - Mandalay Expressway	Initial Pipeline
17	Thailand	ASEAN Digital Hub	Initial Pipeline
18	Thailand	Hat Yai - Sadao Motorway	Initial Pipeline
19	Thailand	Bangkok - Nong Khai HSR - Phase 2	Initial Pipeline
20	Vietnam	Southern Coastal Corridor Project - Phase II	Initial Pipeline
21	Vietnam	Ho Chi Minh City - Moc Bai Expressway	Initial Pipeline

Successful Implementation of Pipeline Project

Below presents the project from initial pipeline that has successfully implemented.

No.	Country Name	Project Name	Initial Pipeline/ Updated Pipeline
1	Expansion of Hang Nadim International Airport	Indonesia	Project has been successfully implemented and would be dropped off from the pipeline

Q&A

Question:

What are the challenges AMS faced while submitting the projects for the pipeline?

Challenges SMBC Observed

- In our engagement with various AMS, the following challenges were highlighted, and we have taken steps to mitigate them.

Challenges	Mitigations
Lack of Bankable Project Submissions	• Ongoing guidance for AMS with provision of our advice to improve bankability of the projects submitted.
Delay in Submission of Project Application Forms	• Abridged Project Application Form was prepared to simplify data collection • Time extension till end April for project sponsors who are in process of preparing submissions.
Discomfort with Sharing Confidential Information	• Initial discussions can omit confidential info • Existing NDA with ASEAN Secretariat ensures secure information handling • Project information will be shared only after project owner’s approval.
Lack of Diversity in Project Submissions	• Proactive engagement with AMS to promote submissions in energy and digital connectivity projects for a comprehensive connectivity strategy.

Suggestions for Project Owners

- The SMBC team is here to assist Project Owners in the development of key selected projects. We have provided some suggestions below for Project Owners to consider.

Choice of projects	Some of the considerations below would be helpful to choose projects that is likely to benefit more from this ASEAN initiative: • Stage of development and type of financing envisioned • Size of project and its impact on the country/region • What is needed for the Project's next stage of development
Preparation	Projects with adequate preparation would set it up for greater chance of success. Some examples of key actionable: • Completed Abridged Project Application Form to provide information for SMBC to evaluate and further advise • Direct engagement of Project Owner with SMBC to discuss the project and for SMBC to provide our advice where needed
Benefits	It would be helpful for project owners to be clear on the type of benefits that they are aiming for. Some of the potential benefits for selected projects are as follow: • Promotion of selected projects to a wider audience – which may include other government bodies, DFIs, MDBs, for **potential collaboration or development funding opportunities** • **Bankability advice and market sounding exercise** for selected projects that are seeking **private financing** • General promotion of project to **increase the profile of the projects** both regionally and internationally

Question:
In your opinion, what does it mean for a project to be bankable?

Poll :
Which factor do you consider the most important for projects to be successful?

Options:

1. Risk Mitigation (Effectively managing and communicating risks associated with the project)
2. Return on Investment (ROI)(Providing competitive and attractive financial returns for investors.)
3. Regulatory Environment (Ensuring a stable and clear regulatory framework that supports investment.)
4. Market Demand (Demonstrating significant demand for the services or infrastructure being developed.)
5. Sustainability and Environmental Impact (Committing to sustainable practices and addressing environmental concerns.)

Informational session: State of the financing market (ECAs, DFIs, commercial lenders) and general bankability considerations

Structured Export & Development Finance

April 2025

SMBC

SMBC x Tomorrow

Rising with Asia
www.smbc.co.jp/asia

Disclaimer

This document has been prepared by Sumitomo Mitsui Banking Corporation (“SMBC”) for the sole purpose of information sharing.

Information and opinions in this document should not be construed as legal, regulatory, financial, investment, tax, or accounting or other technical or professional advice. Recipients should make their own independent appraisal and/or risk assessment of the information and opinions, and consult their professional advisors before embarking on any course of action.

Nothing contained in this document is intended to be, nor should be construed as, a recommendation, solicitation or an offer of any financial products, investments, financing or other banking facilities.

Although the information is obtained from sources considered as reliable by SMBC, SMBC makes no representation or warranty as to, and accepts no liability for any representation in relation to, the accuracy or completeness of the information and opinions contained in this document or for any loss or damage arising from any reliance on or use of this document. SMBC has no responsibility to update any information or opinions contained in this document.

The contents of this document are confidential and shall not be disclosed to any other party without SMBC’s prior written consent. The contents contained in this document may change without prior notice.

Where third party products or services are referred to, they are for information/illustration purposes only and are not a recommendation or endorsement by SMBC. Recipients should make their own independent appraisal of and decision to work with any third party or on the use of their products or services.

SMBC Nikko Securities (Hong Kong) Limited, SMBC Nikko Capital Markets Limited, SMBC Nikko Securities Inc., SMBC Nikko Securities and their affiliates (“SMBC Nikko”) are related companies of SMBC. Information and presentations by SMBC Nikko are made in their own right and SMBC is not authorized to, and does not hold itself out as, representing SMBC Nikko or promoting or entering into products and services for or on behalf of SMBC Nikko. SMBC has not verified the information or materials provided by SMBC Nikko and accepts no responsibility in relation thereto.

Note: Certain images may be subject to copyright. Maps are for illustrative purposes only.
Sumitomo Mitsui Banking Corporation Singapore Branch, incorporated in Japan with limited liability (Reg. No (UEN) T03FC6366F)

Global Agency Coverage

SMBC has an **extensive global platform** which enhances our capability to support our clients who are venturing into new emerging markets and require ECA / multilateral agency funding structures.

Maps shown herein are for illustrative purposes only

Region	EMEA	APAC	AMER
Export Credit Agencies	• Atradius • BpiFrance • Cesce • Credendo • ECIC • EIFO • EKN • Euler Hermes • Finnvera plc • GIEK • OeKB • SACE • SERV • UKEF	• Japan Bank for International Cooperation • Nippon Export and Investment Insurance • Korea Trade Insurance Corporation (K-Sure) • The Export-Import Bank of Korea (KEXIM) • SINOSURE • The Export-Import Bank of China (CEXIM) • Export Finance Australia • Export-Import Bank of Malaysia Bhd • Export-Import Bank of Thailand	• Export Import Bank of the US • Export Development Canada • Seguradora Brasiliera de Credito a Exportacao (SBCE) • Bancomext
Multi-laterals/ DFIs	• BII • CDP • EIB • EBRD • DEG • Finnfund • PROPARCO • FMO • Norfund • Nordic Investment Bank • IsDB • Swedfund	• Asian Development Bank • JOGMEC • JICA • China Development Bank	• World Bank • International Finance Corporation • MIGA • US International DFC • Corporacion Andina Fomento • Inter-American Development Bank (IADB) • Inter-American Investment Corporation (IIC) • Central American Bank for Economic Integration • FinDev Canada

Specialists in Raising Large ECA/Multilateral Financing

SMBC is a **market leader in ECA financing** in Asia Pacific with a **global platform connecting to ECAs & Multilaterals globally**

Project	Debt Volume	ECA	SMBC Roles
Reliance Industries Limited	EUR 302mn & JPY 103bn	SACE	MLA
REC Limited	JPY 60.5bn	SACE	Green Loan MLA
Power Finance Corporation	JPY 25.5bn	JBIC	MLA & Agent
NTPC Ltd & NTPC Renewable Energy Ltd	JPY 30bn	JBIC	MLA and Agent
Olam Treasury Pte Ltd	USD 250mn & JPY 37.38bn	SACE	MLAB, Coordinator& Agent
Balikpapan Refinery	USD 3.1bn	KEXIM, KSURE, SACE, US EXIM	ECA Coordinator & MLA
State Bank of India	JPY 13.5bn	JBIC	MLA
Power Finance Corporation	JPY 2.6bn	JBIC	MLA & Agent
Petrovietnam Power	USD 200mn	SACE	MLA & Agent
Lotte Chemical Indonesia	USD 2.4bn	KEXIM/KSURE	MLA
San Miguel Corporation	USD 2.165 bn	Atradius	ECA Agent, Coordinator, MLA
Rural Power Company Limited	EUR 140mn	Sinosure / Euler Hermes	ECA Coordinator, MLA & Agent
NTPC Ltd.	JPY 50.0bn	JBIC	MLA and Agent
State Bank of India	USD 1.0bn	JBIC	MLA, Coordinator and Agent
PT PLN	USD 500.0mn	MIGA	MLA
Yunlin Offshore Wind	EUR 2.5bn & NTD 85.0bn	EKF / Euler Hermes / Atradius	FA, MLA, Agent for Euler Hermes and Atradius

Source: PFI, IJ Global, The Asset, Inframation, Dealogic, TXF

Top Export Finance Lenders in Asia (2024)				
Rank	**Lenders**	**USD mn**	**#**	**%**
1	**SMBC**	**1,351**	**14**	**8.6**
2	BNPP	1,300	8	8.3
3	HSBC	1,092	11	6.9
3	CACIB	1,092	10	6.9
4	Citi	952	7	6.1
5	ING	841	7	5.3
6	KfW	832	6	5.3
7	Santander	738	5	4.7
8	SCB	677	9	4.3
9	SocGen	655	3	4.2

Source: TXF ECA Finance League Tables 2024

Top Export Finance Lenders in Asia (2023)				
Rank	**Lenders**	**USD mn**	**#**	**%**
1	Mizuho	2,192	14	6.6
2	SocGen	2,127	8	6.4
3	**SMBC**	**1,943**	**21**	**5.8**
4	CACIB	1,826	7	5.5
5	BNPP	1,496	6	4.5
6	Natixis	1,471	5	4.4
7	HSBC	1,442	8	4.3
8	KfW	1,361	4	4.1
9	CIC	1,163	2	3.5
10	Citi	1,099	6	3.3

Source: TXF ECA Finance League Tables 2023

Overview of ECA, DFI, MLA financing

SMBC has **active relationships** with more than 40 agencies across the world. Agency finance solutions can come in the form of corporate loans, project finance loans, and bonds utilizing support from Export Credit Agencies (**ECAs**), Development Finance Institutions (**DFIs**), and Multilateral Agencies (**MLAs**).

	What are they	When to use	Key Considerations
ECAs, among others: Atradius BpiFrance Cesce Credendo ECIC EDC EFA EKF EKN Euler Hermes Finnvera plc JBIC K-Sure KEXIM NEXI OEKB SACE SERV Sinosure UKEF US EXIM	• Government agencies supporting trade and investments to the emerging markets. • Most adhere to an established set of guidelines (OECD Consensus). • Financing is *typically* linked ("tied") to procurement of goods and/or services from the exporting country. Untied financing may be available depending on the ECA.	• *Typically*, when goods and/or services are sourced from the exporting country. • Seeking longer tenors and attractive pricing for capex investments • Optimize usage of client's credit lines	▲ Attractive pricing ▲ Very predictable terms & conditions ▲ Comprehensive cover increase financing appetite from financial institutions ▲ Free up funding sources (bank financing, capital markets) ▼ Rigid eligibility criteria ▼ Strict funding requirements
DFIs & MLAs, among others: ADB BII CDP DEG EIB FinDev Canada Finnfund FMO IsDB IFC JICA JOGMEC MIGA Norfund PROPARCO Swedfund US DFC World Bank	• DFIs are government institutions focusing on developmental projects. They support overseas investments into emerging markets • MLAs are institutions owned by more than one country that support economic and social progress in their member countries • Financing is "untied" in nature	• Project or company results in developmental impact in terms of ESG / UN SDGs. • Agencies are more flexible in the types of transactions they can participate in • More focus on national interest and development impact than national production / trade	▲ Main source of long-term capital in highly volatile environments ▲ Relatively flexible eligibility criteria ▲ Typically longer tenor and competitive pricing relative to commercial financing ▲ Improved market visibility and positive image from ESG perspective ▼ Needs to adhere to certain statutory requirements such as environmental ▼ Fairly extensive execution timeframe

ECA Financing - Overview

SMBC

ECA Financing – Tied vs Untied Financing

Export Credit Agencies (ECAs) are government institutions which support trade and investment by providing financing services such as guarantees, insurance and loans to support domestic companies. ECAs generally have **two types** of funding schemes:

Tied Financing:

- Linked to the value of exports from that country. Use of Proceeds is for payments related to contract(s) with foreign exporter(s) and service provider(s).
- Governed by OECD guidelines for amount, tenor, repayment profile

Untied Financing:

- Borrower must fulfill the eligibilities of respective ECAs' programs (i.e. shareholdings, O&M, offtake of energy/ resources, or future procurement)
- Provides greater flexibility to structure the debt, as repayment profile is not bound by OECD requirements on average life, amounts and tenor.

	Tied Financing	Untied Financing
Amount	85% of the value of exports from country of origin. A discretionary allowance for local content of up to 40% of export value (Cat 0 countries) or 50% (emerging markets).	Flexible, not linked to value of exports, subject to ECA approval.
Availability Period	To match construction / procurement	To match needs
Repayment Period	Maximum repayment tenor post completion under OECD Consensus varies depending on location and nature of underlying project and financing structure. Longer tenors up to 18 years can be possible for some sectors.	Flexible, not governed by OECD guidelines
Type of Cover	Comprehensive Guarantee/Insurance Cover	Comprehensive Guarantee/Insurance Cover or Extended Political Risk Cover
Coverage	Depending on ECA and project risk assessment: Political Risk: Up to [100]% Commercial Risk : Up to [100]%	Depending on ECA and project risk assessment: Political Risk: Up to [100]% Commercial Risk : Up to [100]%
Premium Payment	Typically upfront; may be financeable or payable as per disbursement schedule subject to ECA approval	Typically upfront; may be financeable subject to ECA approval

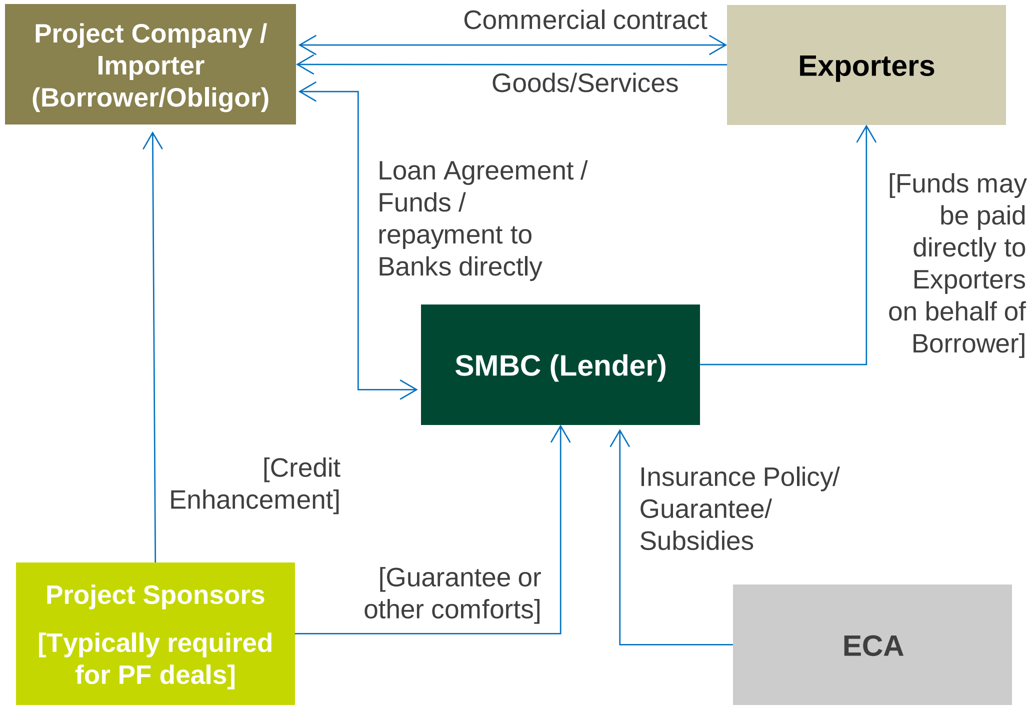
ECA Tied Financing – Overview

- The level and nature of ECA involvement will be **driven by the EPC and equipment procurement** strategy (in case of Buyers' Credit financing programmes).
- Able to provide **large liquidity at long tenors** and **relatively competitive pricing** which is based on highly rated agency risk
- **Increased market appetite** for the financing as it does not count against client bank lines given ECA cover.
- Can be done on corporate finance (**CF**) or project finance (**PF**) basis.
- **Environmental and social issues** will be a key area of due diligence by the ECAs.
- Tied Financing would be **governed by OECD guidelines** for amount, tenor, repayment profile, whereas there is greater flexibility with Untied Financing.

ECAs - Tied Financing General Terms	
General	Linked to the value of exports from that particular country. Governed by OECD guidelines for amount, tenor, repayment profile
Financing Amount	85% of the value of exports from country of origin, plus up to [50%] of export value for local content
Repayment Period	Repayment period up to 15 years in addition to construction / delivery period however not to exceed the useful life of the goods/services. Longer tenors may be considered depending on the purpose of financing*
Average Repayment Life	Not more than the greater of 65% of repayment period or 6 years
Type of Cover	Comprehensive Guarantee/Insurance Cover
Coverage	Depending on ECA, but could be up to [100]% for political risk and up to [95-100]% for commercial risk
Premium Payment	Upfront or annually (most of the ECAs require upfront payments which can be included in the financing in some cases)

** Climate change mitigation sectors (e.g. renewable energy, sustainable transport etc. may benefit from a longer repayment term on a case-by-case basis under new OECD guidelines released in 2023*

Typical Buyer's Credit Tied Financing Transaction Structure



Structures shown herein are for illustrative purposes only

ECA Untied Financing – Overview

- Untied programs provide **greater flexibility** as the proceeds are **not linked to export contracts** and terms are not bound by OECD guidelines.
- **Relatively competitive pricing** which is based on highly rated agency risk
- **Increased market appetite** for the financing as it does not count against client bank lines given ECA cover.
- Amounts, tenor, and eligibility is dependent on the respective ECA's programs which may be unique and assessed on a **case-by-case basis**.

Selected ECA untied programs are shown below:

ECA	SACE (Italy)		JBIC (Japan)	NEXI (Japan)	K-Sure/ KEXIM (S. Korea)	EFA (Australia)
Program	Push Strategy	Strategic Import Financing	Untied Loan / JBIC GREEN and Overseas Investment Loan	Overseas Untied Loan Insurance (OULI) / LEAD	Overseas Business Financing Insurance/ Guarantee	Infrastructure / Southeast Asia Mandate
Eligibility	Sovereign, SOEs, and Corporates. Potential foreign buyers of Italian goods & services in strategic sectors.	Large foreign suppliers of strategic goods with undertaking to increase sales to Italy.	SOEs and corporates where the project involves Japanese sponsor/company or green projects where the company has utilised Japanese technologies.	Sovereigns, corporates / Projects with Japanese involvement. Focus on leading technologies, environment, alliances, and development. (LEAD)	Overseas projects with Korean interests (i.e., Korean offtaker, investor, operator, etc.)	Infrastructure projects in Indo-Pacific and Southeast Asia region. Infrastructure includes telecoms, energy, electrification, water, sanitation and transportation
Cover	Up to 80%		Up to 90 – 95%	Up to 90 – 100%	Up to 100%	Up to 95%
Tenor	Up to 12 years		Dependent on purpose	Dependent on purpose	Dependent on purpose	Dependent on purpose
Comments	**PROS**: •Generally flexible. Push program only requires best-efforts procurement undertaking. •Long tenor up to 12 years possible **CONS**: •Lower credit rating compared to other ECAs •Slightly longer documentation process than standard commercial facilities		**PROS**: •Long tenor and competitive pricing •Can provide direct loan and guarantee •**CONS**: •Require some Japanese company involvement •May require a longer due diligence and documentation process	**PROS**: •Flexible on Japanese interest •Long tenor and competitive premiums **CONS**: •Upfront payment of premium •May require a longer due diligence and documentation process	**PROS**: •Long tenor and competitive pricing **CONS**: •Require Korean company involvement •May require a longer due diligence and documentation process	**PROS**: •Long tenor and competitive pricing •Can provide direct loan and guarantee •**CONS**: •May require Australian angle •Amount could be limited •May require a longer due diligence and documentation process

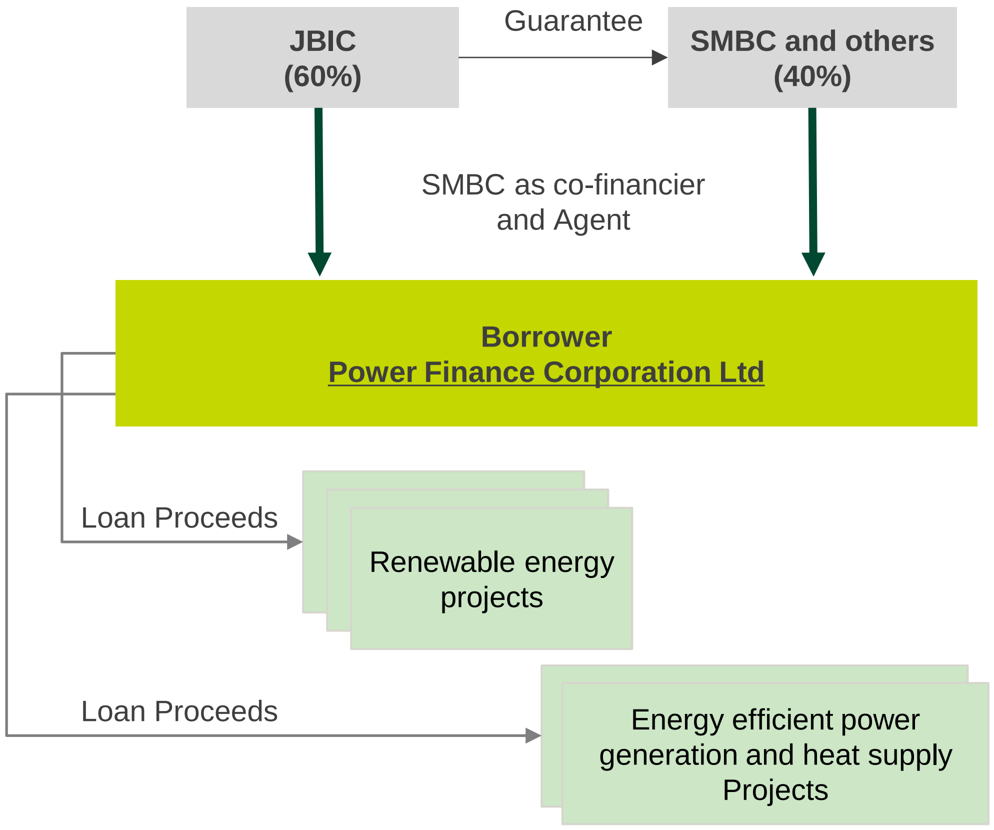
JBIC GREEN– Power Finance Corporation ("PFC")

On 7 July 2022, JBIC signed a General Agreement for credit line totalling up to JPY 30bn (of which JBIC's portion is JPY18 bn) with PFC, a government financial institution in India.

Project Highlights

- ✓ PFC, the provider of approximately 25% of the financing owed by the entire power sector of India, is an important institution for realizing the electricity policy of the Government of India. The credit line will support the initiatives that the Government of India promotes for the purpose of global environmental preservation.
- ✓ The General Agreement is an umbrella facility extended under JBIC's GREEN operations and is intended to provide funding through PFC for renewable energy projects and energy efficient power generation and heat supply projects in India.
- ✓ Under the General Agreement, JBIC subsequently signed three loan agreements to support PFC's on-lending to a wind farm project and two waste-to-energy projects in India.
- ✓ SMBC acted as co-financier and agent of the financing.

Transaction Structure



Source: JBIC Website

Structures shown herein are for illustrative purposes only

DFI/ MLA Financing

SMBC

DFI /MLA Financing – Overview

- While each DFIs / MLAs may have specific focus areas, they generally provide financing to **development finance projects or companies aligned with the UN Sustainable Development Goals (UN SDGs).** Examples includes Green (e.g. renewable energy), Social (e.g. MSMEs, affordable housing, women-focus), or Blue projects (e.g. clean water)
- Able to provide **financing for longer tenors** at **relatively competitive pricing** (some DFIs offer concessional funding as well including through grants/TA)
- Can be done on corporate finance (**CF**) or project finance (**PF**) basis depending on the agency
- **Environmental and social issues**, and **developmental impact** will be a key area of due diligence by the DFI / MLA
- SMBC can work with various DFIs / MLAs to coordinate a co-financing package to achieve the overall financing requirements

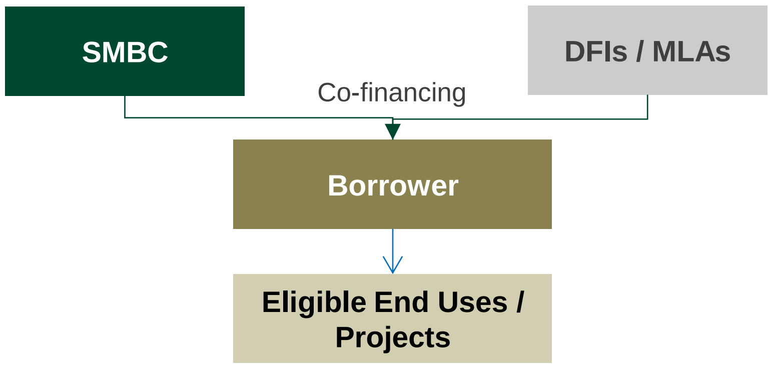
DFI / MLA - Co-financing General Terms*	
General	Structured as two tranches consisting of a DFI/MLA tranche and a commercial bank tranche. DFI/MLA tranche will require eligible use of proceeds in line with UN SDGs (e.g. Green / Blue / Social impact).
Financing Amount	Dependent on size of project / eligible use of proceeds. Typically, up to 70% of project cost or around USD [10-150]mn (per DFI/MLA)
Tenor	DFI / MLA tranche: generally, longer than commercial banks Commercial tranche: in line with other loans from commercial banks As a whole, the blended average life and tenor of the facility will be longer than typical commercial loans.
Repayment	Case-by-case basis, typically semi-annual repayment.
Currency	Depending on DFI/MLA, JPY, USD, EUR, local currencies (long-term funding is possible if swap is available in the market)
Pricing	Blended pricing of the facility is typically competitive compared to other commercial loans. Certain DFIs/MLAs may be tax exempt and may additionally provide technical assistance (TA) and grants.

** This represents broad terms generally seen and it may differ for each DFI/MLA.*

SMBC has a wide network with DFIs & MLAs including:

DFIs	JICA, FinDev Canada, Proparco, DEG, CDP, BII, US DFC, Finnfund, FMO, Norfund, Swedfund
MLAs	ADB, World Bank, MIGA, IFC, EIB, EBRD, Nordic Investment Bank, IsDB, CAF, IADB, IIC, CABEI

Typical Transaction Structure:



Structures shown herein are for illustrative purposes only

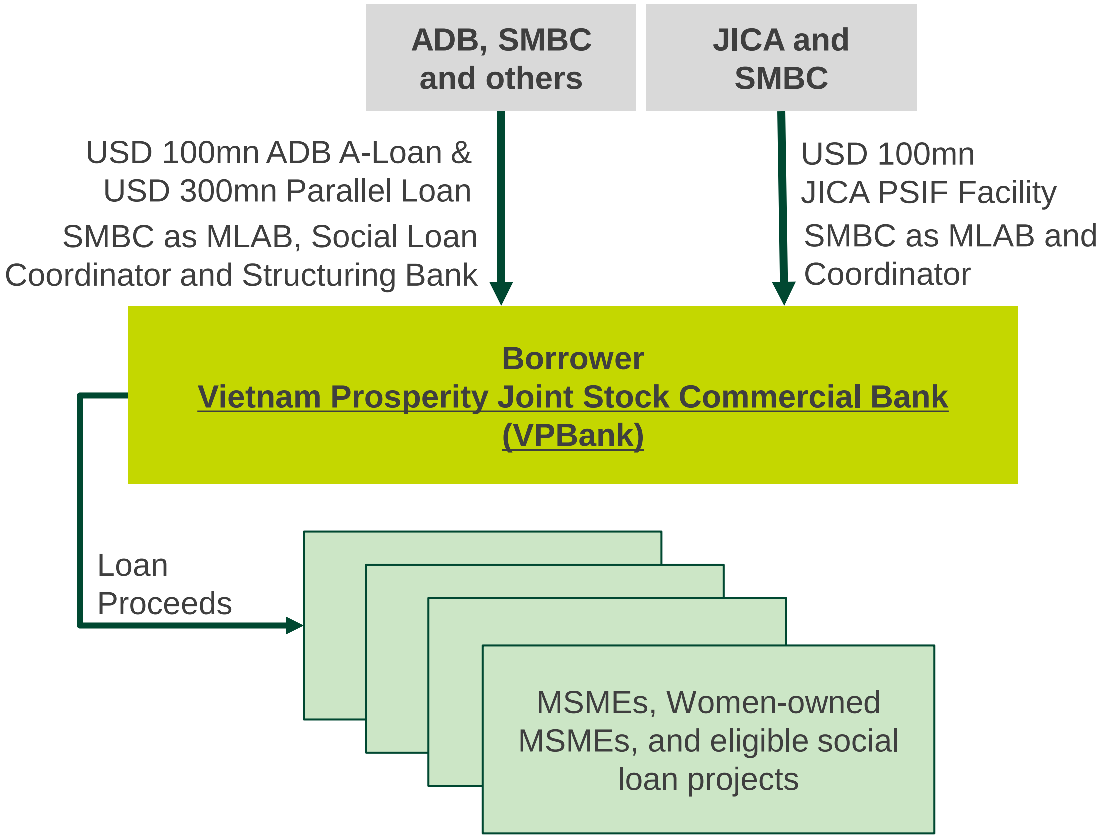
Case Study – Vietnam Prosperity JSC Bank (VPBank)

SMBC supported VPBank in raising a total of USD 500mn consisting of three facilities involving ADB, JICA and other commercial lenders to support VPBank's further growth in financing MSMEs in Vietnam.

Project Highlight

- ✓ Vietnam Prosperity Joint Stock Commercial Bank ("VP Bank") is a commercial bank in Vietnam established in 1993 with a focus on financing to Micro, Small and Medium Enterprises (MSMEs), consumers and the retail sector, with more than 100,000 MSME customers in Vietnam.
- ✓ SMBC supported VPBank in closing a co-financing facility with JICA signed in October 2021, an ADB A loan and a parallel social loan signed in November 2022.
- ✓ The facility will help VPBank expand lending to MSMEs out of which a certain percentage of proceeds must be used for loans towards women-owned MSMEs. This will contribute to Goals 5, 8, 9, and 17 of the United Nations' Sustainable Development Goals (SDGs).
- ✓ SMBC acted as <u>JICA Coordinator, Social Loan Coordinator, Structuring Bank, MLAB and Facility Agent.</u>

Transaction Structure



Source: JICA, ADB and VPBank Websites

Structures shown herein are for illustrative purposes only

Overall Financing Plan & Role of the Coordinator

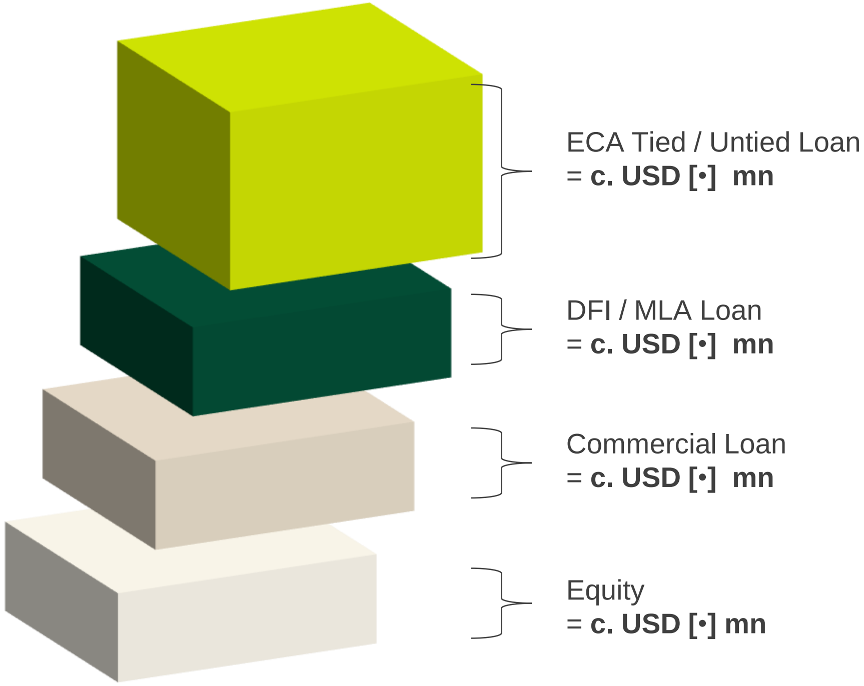
SMBC

Potential Financing Plan

For large projects with **substantial financing requirements**, a **blended finance approach** may be optimal involving a **mix of financing** from ECAs, DFIs, MLAs and commercial lenders.

Overall financing plan:

- A blended finance approach can help to fully leverage the benefits of large liquidity, long tenor and competitive pricing.
- Consider project specifics to size the respective tranches accordingly



Total Debt + Equity = c. USD [•] bn

Potential Considerations

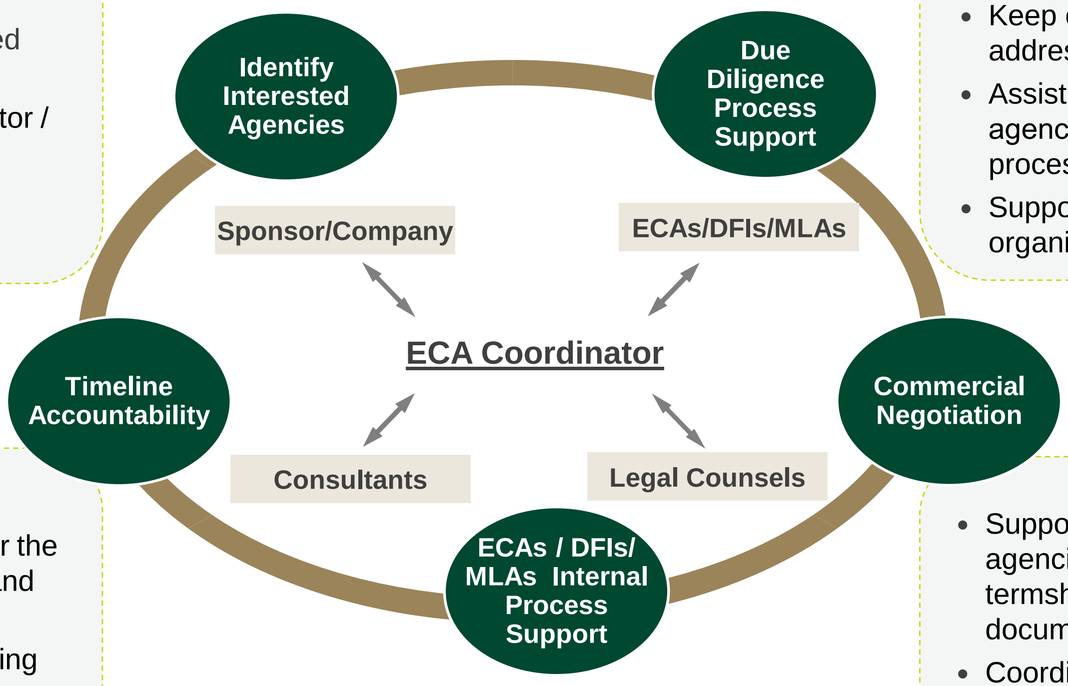
1. **Project type & structure –** What is the underlying asset/purpose and if agencies can support? Is project finance being considered? Any security to be provided?
2. **Overall financing amount, tenor, repayment structure** – any requirements on tenor, repayment structure that is not typically covered by typical corporate financing?
3. **Borrowing entity –** Is the Borrower a corporate, SOE, sovereign or bank? Will a two-step structure be considered?
4. **EPC / Supplier / Procurement strategy –** Is there clarity on procurement strategy or will financing be required beforehand?
5. **ESG / Development Impact –** Is there an ESG angle to consider which may be relevant for the agencies?
6. **Financing timeline –** Each tranche may be put in place separately which can help in meeting your financing timeline requirements.

Role of a Coordinator

Through SMBC's **sector expertise** and **wide experience** working with various ECAs, DFIs, MLAs, SMBC can support you in coordinating a **robust and optimal** financing package with the relevant agencies to ultimately achieve a timely financial close.

- Identify ECA eligibility for tied or untied ECA financing (if relevant).
- Identify DFIs / MLAs with aligned sector / country interest.

- Support the Company and the involved agencies in their selection of the agencies' consultants/ advisors.
- Keep close contact with the agencies to address any issues as they arise.
- Assist the Company in navigating the agencies' due diligence and approval process.
- Support the involved agencies to organize the due diligence as required.



- Developing a deliverable timetable for the ECAs/DFIs/MLAs approval process and setting out key steps and milestones.
- Ensuring timely approvals and providing regular progress updates to the Company.

- Supporting the company and the involved agencies during negotiation of the termsheet and the financing documentation.
- Coordinating with the company and agencies to facilitate the resolution of key commercial issue.

Coffee Break
(15 mins)

Question:
If you can choose one infrastructure project, what kind of project do you think your country need the most right now?

Question:
What benefits does your country anticipate from including its proposed infrastructure project in the pipeline?

Presentations by project owners on potential projects

Development of an ASEAN Space Data Infrastructure (Philippines)

Hatyai – Sadao Motorway
&
Highway connecting Udon Thani – Bueng Kan

(Thailand)

Lunch Break

(Workshop will resume at 13:30pm)

Question:
What is your country's most significant landmark/ infrastructure?

Informational session:
Key developments, challenges, and opportunities in transportation infrastructure in the region

Finance in Infrastructure Sector

April 2025

SMBC

SMBC x Tomorrow

Rising with Asia
www.smbc.co.jp/asia

Disclaimer

This document has been prepared by Sumitomo Mitsui Banking Corporation (“SMBC”) for the sole purpose of information sharing.

Information and opinions in this document should not be construed as legal, regulatory, financial, investment, tax, or accounting or other technical or professional advice. Recipients should make their own independent appraisal and/or risk assessment of the information and opinions, and consult their professional advisors before embarking on any course of action.

Nothing contained in this document is intended to be, nor should be construed as, a recommendation, solicitation or an offer of any financial products, investments, financing or other banking facilities.

Although the information is obtained from sources considered as reliable by SMBC, SMBC makes no representation or warranty as to, and accepts no liability for any representation in relation to, the accuracy or completeness of the information and opinions contained in this document or for any loss or damage arising from any reliance on or use of this document. SMBC has no responsibility to update any information or opinions contained in this document.

The contents of this document are confidential and shall not be disclosed to any other party without SMBC’s prior written consent. The contents contained in this document may change without prior notice.

Where third party products or services are referred to, they are for information/illustration purposes only and are not a recommendation or endorsement by SMBC. Recipients should make their own independent appraisal of and decision to work with any third party or on the use of their products or services.

SMBC Nikko Securities (Hong Kong) Limited, SMBC Nikko Capital Markets Limited, SMBC Nikko Securities Inc., SMBC Nikko Securities and their affiliates (“SMBC Nikko”) are related companies of SMBC. Information and presentations by SMBC Nikko are made in their own right and SMBC is not authorized to, and does not hold itself out as, representing SMBC Nikko or promoting or entering into products and services for or on behalf of SMBC Nikko. SMBC has not verified the information or materials provided by SMBC Nikko and accepts no responsibility in relation thereto.

Note: Certain images may be subject to copyright. Maps are for illustrative purposes only.
Sumitomo Mitsui Banking Corporation Singapore Branch, incorporated in Japan with limited liability (Reg. No (UEN) T03FC6366F)

Table of Contents

Source: Inframation, PFI

Introduction

01

Introduction

Transport plays a key role in economic development with increasing demand for connectivity, urban transit and green solutions.

Traditional role of Transportation

- Providing access to goods and basic services.
- Promoting regional integration and facilitating trade.

Asia's transport infrastructure has seen substantial improvements over time, though significant gaps persist.

For most transport sub-sectors, Asia continues to lag behind developed countries in infrastructure provision.

- Differing rate of population growth, economic development and urbanization have resulted in significant variations in infrastructure stocks across the Asia-Pacific region.
- While governments in Asia are already investing in transport infrastructure to expand the capacity of the systems, additional private sector capital is required to accelerate project delivery.
- Governments across the region have become cautious about further borrowing to fund infrastructure expenditure and are increasingly adopting alternative methods to finance infrastructure projects.
- Increased private sector capital / debt will require more certainty on returns, higher project quality, longer period for project preparation and stronger governance.
- Although large institutional investors are becoming more prepared to invest in Asian infrastructure, a number of regulatory, political and institutional challenges remain.

How much investment in transport Infrastructure will be made in Asia in the next decade?

Presentation Objective: Discuss trends, challenges, and opportunities in the Transportation Sector in Asia

Transportation Infrastructure Overview

02

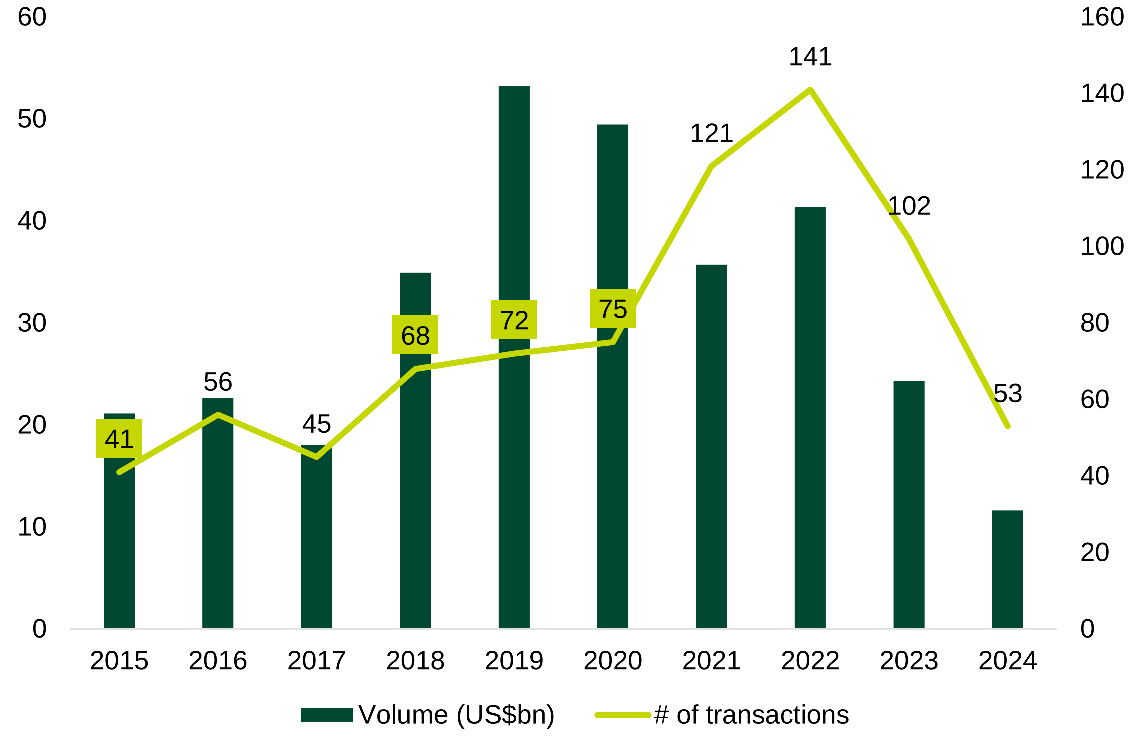
Overview of Transport Sector in Asia

Transport Infrastructure Activities have been robust over the past decade, with notable activity observed in India

Transaction Volume (US$bn) & Count (#) in the Transport Sector

53
Closed transactions in 2024

US$11.62bn
Volume of Closed transactions in 2024



Key Transportation Infrastructure Activity in Asia & Observations

Toll Roads

- Recent Toll road projects are predominantly located in India
- The market has seen an increase in the cost of debt for Toll Road Projects.

Rail

- Recent Rolling Stock projects are predominantly located in Vietnam and China

Ports

- Recent Port projects are predominantly located in India and Thailand
- The market has seen an increase in the cost of debt for Port Projects. This observation is seen alongside the declining growth prospects of such projects.

Airports

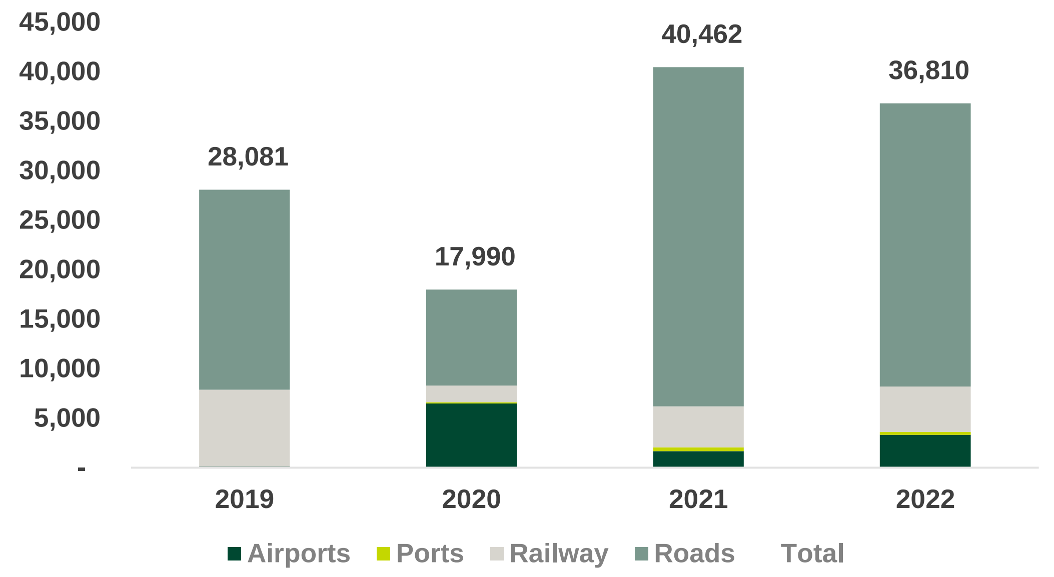
- Recent Airport projects are predominantly located in India and China
- The market has seen an increase in the cost of debt for Airport Projects. This observation is seen alongside the growing prospects of such projects.

Source: Fitch Ratings, Infralogic

Overview of Asia Transport Infrastructure Investment

Private participation contributes a large portion to the development of Transport Infrastructure in Asia

Private Participation Investment in Asia (US$m)



- Private participation in Asia Transport Infrastructure has generally increased by 31% from 2019-2022.
- Road infrastructure assets continued to receive the highest level of private investment, followed by airport sub-sector.
- Between 2019 and 2022, there was a notable shift towards investment in airports and ports, with the total investment in these two subsectors rising more than 30 times during this period.

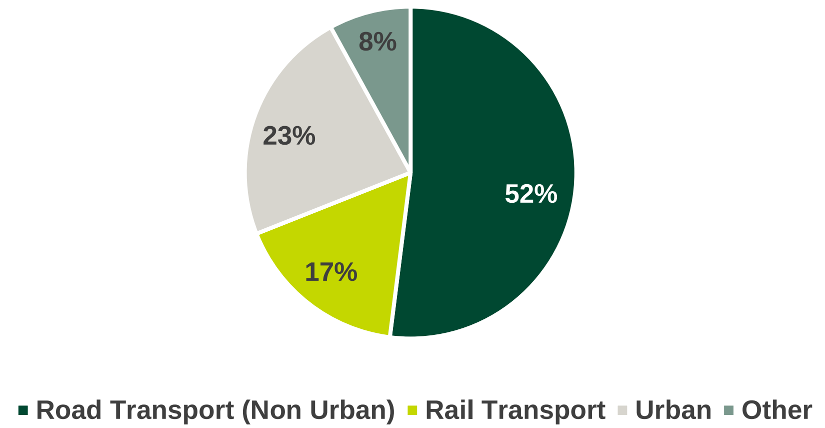
MDB Investment in Asia (2020-2024)

US$4.6bn

Average annual loans and grants towards transport investments

21%

Proportion of loans and grants issued by ADB towards transport investments



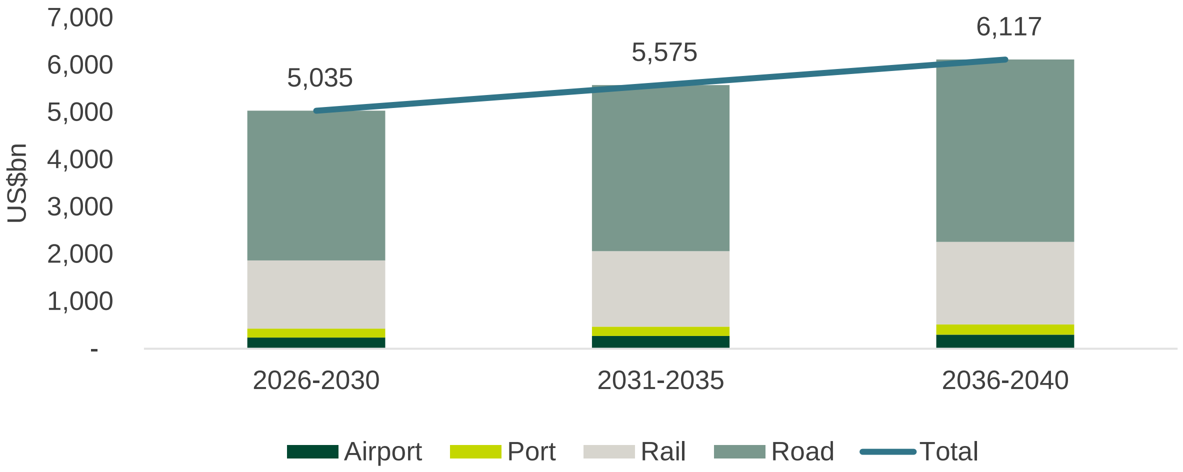
- Road transport represents the largest and most prioritized investment in Asia for MDBs.
- Bilateral Development Agencies have traditionally focused more on rail (e.g. JICA) or road and urban rail (e.g. CEXIM)
- However, transport infrastructure in Asia is lower than would be expected, with road infrastructure only accounting for 30% of the global road total, and rail infrastructure for 34%.

Source: ADB, PPI

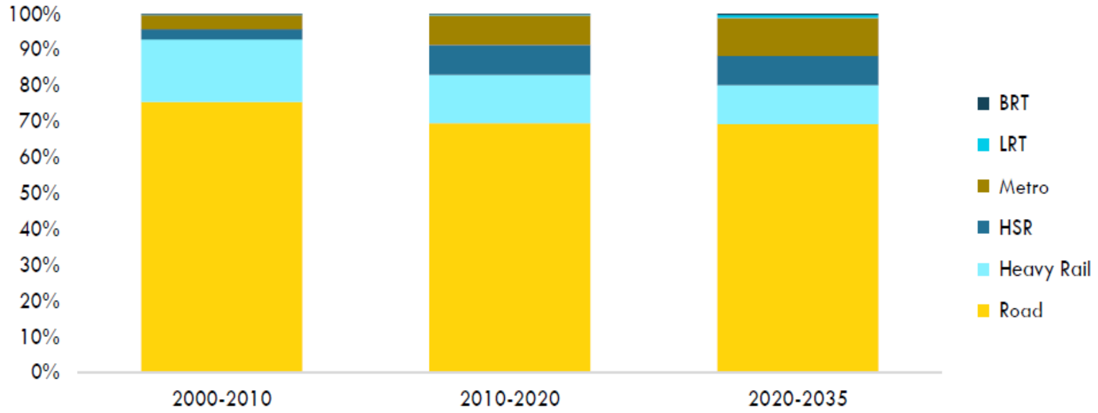
Significant Financing Requirement in Asia

Asia offers an opportunistic infrastructure market with investment needs estimated over US$10tn over the next 10 years which requires proactive planning and resource mobilization

Infrastructure – ~US$ 17.5tn over 15 years



Transport Infrastructure Investment by mode

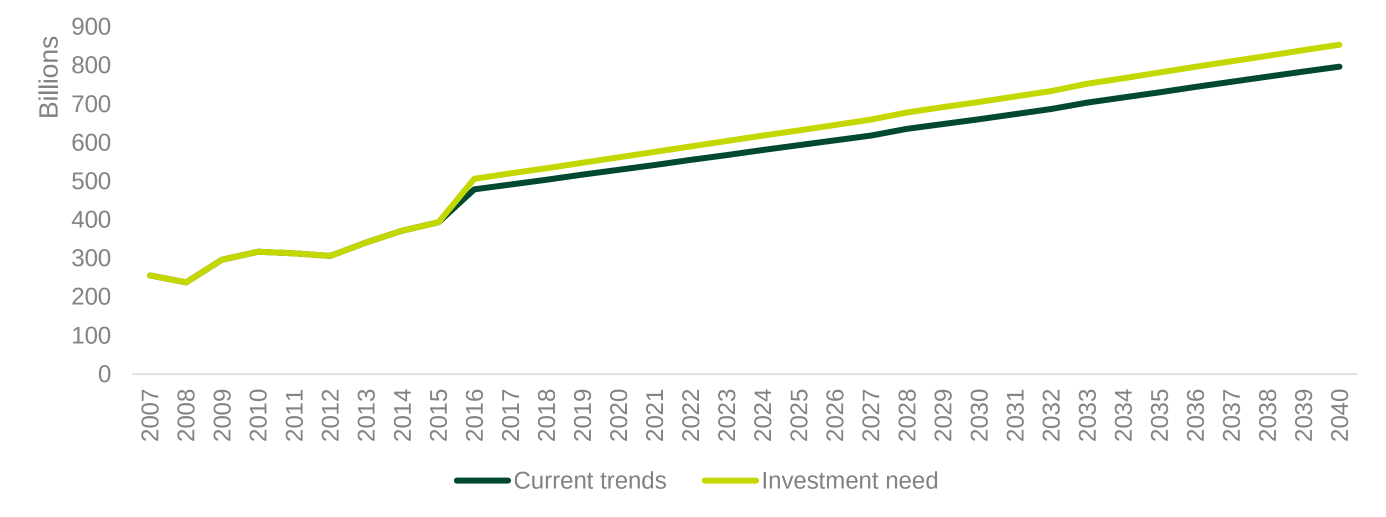


- Overall investment need, over the next 15 years (2026 – 2040), in Asia's infrastructure sector (roads, rails, airports, and ports) is estimated at US$17.5tn.
- The overall structure of infrastructure spending is forecasted to remain consistent with modest investment shifts.
- Roads will continue to command the largest share of investments for the next 10-15 years to accommodate the anticipated vehicle ownership growth and enhance connectivity, followed by Rail and Airport.
- Growth across each sub-sector is expected to be driven primarily by upper-middle income economies followed by lower and lower-middle income.

Source: Global Infrastructure Hub; International Energy Agency (IEA), Infralogic

Toll Roads

Stable growth is expected amid robust economic growth, strong domestic demand and inelastic commuter traffic



US$16tn
Current investment (Asia) in Roads

US$17tn
Total investment needed

US$1tn
Investment gap

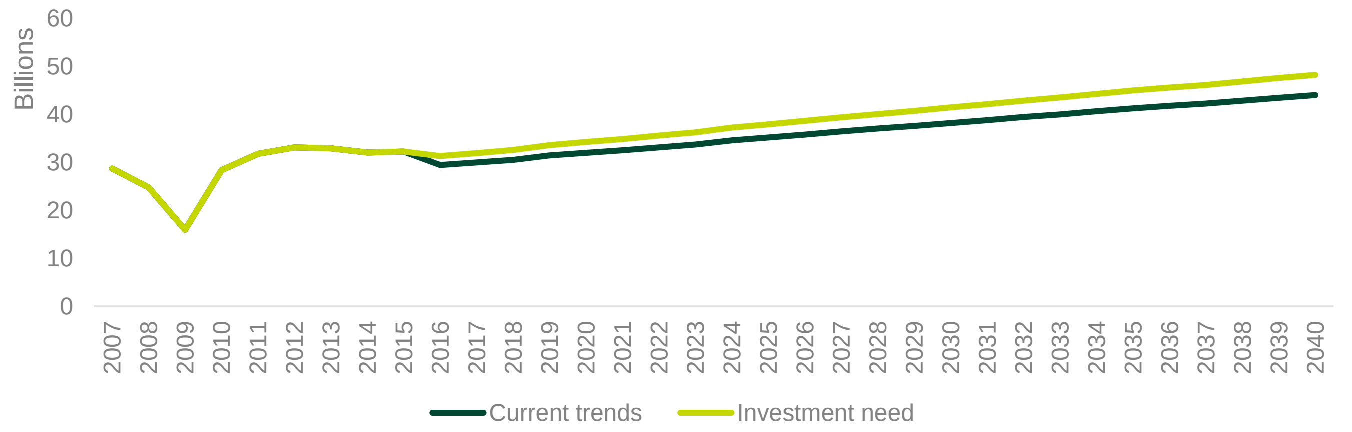
Demand Drivers:

Fundamentals Support Capex Needs	▪ Improved transportation infrastructure will support GDP growth in APAC and boost demand for better toll-road connectivity. ▪ This will drive significant investments in toll-road expansion and upgrades to reduce congestion and travel times. ▪ Easing input costs should make brownfield projects more feasible, while greenfield projects may face delays and cost overruns due to complexity.
Increasing Private Sector Interest	▪ Interest in toll-road projects is increasing from private sectors, pension funds, and sovereign wealth funds, focusing on brownfield assets with stable revenue streams. ▪ Toll road investments through InvITs are gaining popularity in India due to benefits like leverage caps and restrictions on under-construction assets, reducing risks from delays and aggressive financial strategies. InvITs also typically include diversified, operationally proven projects.
Robust Financial Profile	▪ Toll-road projects are expected to maintain strong financial profiles, supported by solid traffic volumes. ▪ Acquisitions are driving growth, with operators targeting EBITDA-enhancing assets with proven track records. ▪ While high capex and debt may strain leverage short-term, long-term benefits include extended concession life and steady EBITDA growth.

Source: Fitch Ratings, Global Infrastructure Hub

Ports

Ongoing geopolitical tensions and trade disputes are expected to foster trade protectionism, with countries increasingly fortifying their supply chains, shifting trade dynamics



US$923bn

Current investment (Asia) in Airports

US$998bn

Total investment needed

US$76bn

Investment gap

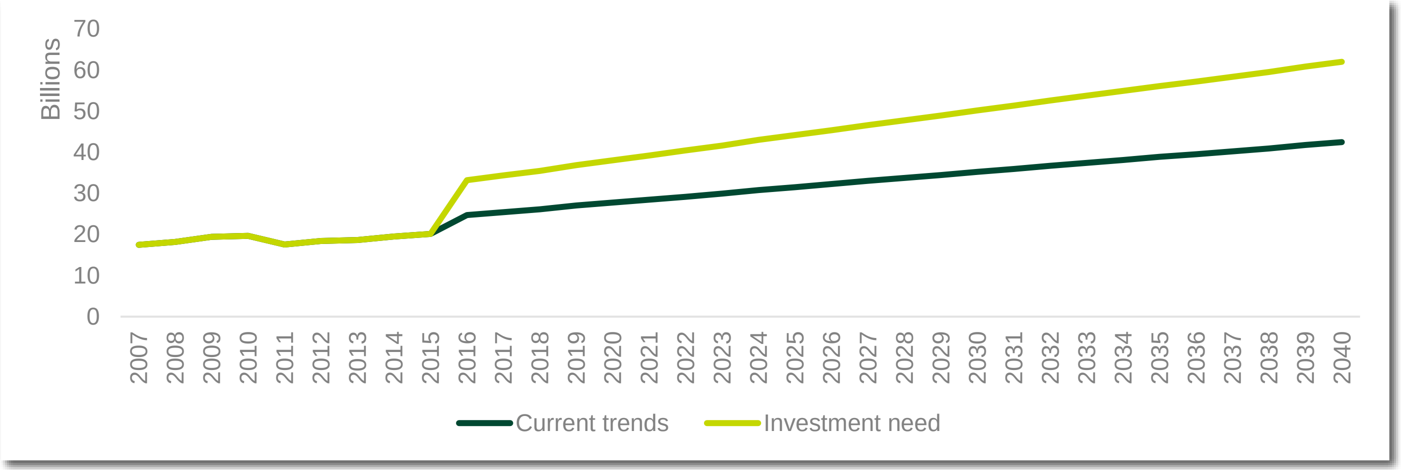
Demand Drivers:

Trade Rebalancing Support Volume	▪ Volume growth at APAC seaports to stay steady, supported by stabilizing GDP growth, the region's economic resilience and the strategic positioning of key countries. ▪ The rebalancing of supply chains should benefit Vietnam, India, Thailand, Indonesia and Malaysia as companies diversify and secure their supply routes. These countries are likely to see increased trade volume, boosting port activity.
Capex and Consolidation Accelerates	▪ Resilient volume amid robust GDP growth should lead to a further uptick in expansionary capex. ▪ Increasing containerisation in countries like India and a surge in transshipment volume has prompted higher capex activity, including for the development of greenfield ports. Some APAC countries are also experiencing industry consolidation aimed at gaining scale and operational efficiency. ▪ Post-consolidation capex to enhance synergies and operational efficiency to drive investment in countries like Indonesia.
Resilient Cash Flow Profiles	▪ Stable volume and easing cost pressure have strengthen APAC seaports' financial profiles and have resulted in less volatile cash flow. ▪ Leverage expected to remain low for the medium term. This will allow port operators to pursue more ambitious capex, while some regions, like India and Indonesia, are likely to form merger plans to acquire high-quality assets and strengthening their competitive positions. ▪ Most operators have flexibility with the timing of capex plans, with lower debt funding costs further enhancing financial profiles.

Source: Fitch Ratings, Global Infrastructure Hub

Airports

Air traffic growth in APAC is expected to remain robust on resilient travel demand, with leisure travel and short- haul flights to remain the key growth drivers



US$841bn
Current investment (Asia) in Airports

US$1.2tn
Total investment needed

US$350bn
Investment gap

Demand Drivers:

Expanding Domestic Travel	▪ The robust air travel demand is driven by expanding domestic travel and airlines extending routes. ▪ Significant infrastructure investments and the proliferation of low-cost carriers also enhance market access, particularly in underserved areas. These factors ensure long-term growth and strong performance for regional airports.
Rising Non-Aero and Cargo Revenue	▪ Non-aero and cargo revenue is expected to surge in 2025, driven by strong air travel demand and large investments in airport logistics infrastructure. Operators aim to maximise returns by leveraging the strategic city locations of airports to develop comprehensive retail hubs. ▪ Ongoing geopolitical conflict will boost air cargo volume as shippers seek alternatives to maritime routes.
Growth Expectations Drive Investments	▪ Continued high capex at APAC airports, driven by strong long-term growth fundamentals, with investments focused on expanding capacity through greenfield and brownfield projects. ▪ Factors such as robust GDP growth, a rising middle class, and low-cost carriers will further increase air travel demand. ▪ Surge in air travel attracting private sector interest in airport infrastructures.

Source: Fitch Ratings, Global Infrastructure Hub

How to Enhance Private Sector Participation in Infrastructure Development

03

PPP in Asia Pacific

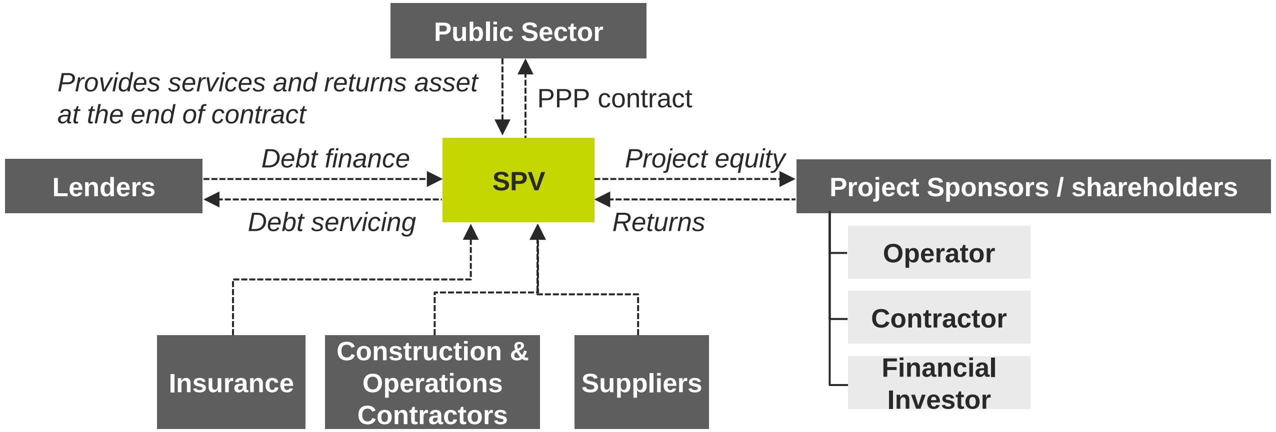
Infrastructure investment across the AP region has historically been undertaken almost exclusively by governments with little private sector involvement. There is growing recognition that PPPs have a vital role to play in bridging the infrastructure gap across Asia.

Sources of Financing (PPI database as of 2024)

Overview of PPPs

- PPPs offer access to private capital which alleviate the fiscal burden on governments by spreading the upfront capital cost of infrastructure over the assets' lifetime and also allow governments to tap into private expertise to deliver and operate complex projects
- In APAC, PPP can refer to a wide spectrum of procurement models to describe private sector participation in public service delivery including Design & Build, O&M, DBO, DBFO, Build-own-operate, etc.
- In APAC, number of PPP projects has grown by 35% between 1995 to 2023, with total investment amounting to US$911bn as of 2023. Of which, greenfield projects account for c.80% of projects in APAC by investment value.

Typical PPP Structure



Challenges and Benefits of PPP

Challenges of PPPs

- Governance
- Institutional structure and capacity constraints
- Weak PPP laws and policies
- Weak country and sovereign risk ratings

Benefits of PPPs

- Increased bankability
- Revenue sharing and BOT agreement allows private operator to reinvest initial earnings without the immediate pressure of sharing profits
- Efficient operation and management with private players having a stake in the infrastructure projects

Sources: Global Infrastructure Hub, Global Infrastructure Outlook, PWC, BCG, PPI Database, PwC, BNEF, IRENA

Infrastructure Development Lifecycle

Bankability assessment is critical early on to attract investors and lenders and to ensure smooth execution of the Project

- Transportation infrastructure projects require level of sector expertise, long implementation periods and associated risks.
- Thus, it is thata project preparation agency addressed the riska involved in the most transparent and detailed way possible
- Financing considerations start at project conceptualization and evolve across all phases.
- Construction risk mitigation, cost overrun protection, interest and FX risk hedging, independent monitoring.
- Revenue stability post-construction.
- Refinancing and asset monetization help optimize capital efficiency.

Phase	**Planning**	**Preparation**	**Procurement and Financing**	**Implementation**

Tasks	▪ Needs Identification
	▪ Sustainable Infrastructure plan
	▪ Infrastructure pipeline
	▪ Project selection
		▪ Feasibility Studies
		  ▪ Technical
		  ▪ Economic
		  ▪ Environment
		  ▪ Social
		  ▪ Legal/regulatory
		▪ WLCC analysis
		▪ Government Funding
			▪ Select procurement strategy
			  ▪ Public procurement
			  ▪ PPP
			▪ Select financing strategy
			▪ Contractual efficiency
				▪ Design
				▪ Construction
				▪ Operation
				▪ Maintenance
				▪ Disposal
				▪ Monitoring and Evaluation

Decision	**Investment decision**	**Validation of Investment decision**	**Procurement and financing strategy decision**	**Asset management strategy decision**

Source: ESCAP

Role of Banks and Financial Institutions

Banks and Financials Institutions help bridge gaps which help the infrastructure sector to overcome investment challenges

Infrastructure Sector ⟷ **Investment Challenges**

Role of Banks and Financial Institutions

- Clearly defining a set of priorities for the transport infrastructure sector, and adopting a framework to allow effective project selection.
- The set of priorities should include modal and cross-border connectivity, with considerations for environmental and social sustainability.
- Placing a strong emphasis on economic viability and commercial discipline, to ensure financial sustainability of projects and to crowd in more private capital;
- Working effectively with partners (MDBs, other developmental agencies, private sector and think tanks).
- Early participation of financiers during project preparation can provide greater confidence around funding and improve the conditions for projects to reach financial closure successfully.

Key Bankability Considerations

04

Key Bankability Considerations

- Governments around the world have continuously explore effective ways to deliver the needed infrastructure assets through their respective PPP programs in the face of limited budgetary resources and targeted risk allocation.
- PPP programs need to be attractive to private investment and financing by channeling public sector resources while limiting the budgetary allocation to the relevant infrastructure projects.
- Prospective lenders will need to be comfortable with a clear risk allocation between the public and the private sectors and the available mitigants for the various project risks.
- Below are some example of financial viability support mechanisms that have been used in other countries for capital intensive projects with sizeable revenue (demand or price) risk. These concepts can be varied and/or combined to suit specific needs of a particular project / country.

	Description
Construction Grant / Viability Gap Funding	▪ Usually effective in capital-intensive projects with high front-end costs. The grant will be part of the source of financing which would reduce the overall funding requirement. ▪ There can also be debt caps put in place, again to reduce/control gearing. ▪ By bringing down the debt requirement it would result in improving the Debt Service Cover Ratio (“DSCR”) for lenders and also enabling the break evens to be sufficiently high enough in the early stage to make it bankable. ▪ Higher coverage ratios could also enable the borrower to re-leverage at the later stage when the asset matures.
Government Capital Contribution	▪ Unlike the construction grant, the capital contribution would be injected upon completion or certain construction milestone being achieved. ▪ This cash injection by the government would enable the Borrower to be able to obtain separate quasi-government guaranteed tranche financing which would be more competitive than the standard project finance term loan. ▪ The project would be able to reduce its leverage post completion.

70

Key Bankability Considerations

	Consideration
Minimum Revenue Guarantee ("MRG") (from government or project promoter)	▪ Usually structured as guarantees to cover the project's revenue shortfalls to a certain guaranteed level in order to reduce the volatility in project revenues. ▪ Provides greater certainty on the cash flows and partial mitigation of the market/traffic risk as government/promoter shares the demand or revenue risk. ▪ The overall cost of private funding is expected to be lower as the government/promoter takes on more risk. ▪ Can structure a reset of the Minimum Revenue Guarantee (e.g. after 10 years of operations) once the project stabilizes.
Operation Grant	▪ Supplement project revenues and bring down the cash flow volatility to meet the operational expenses and meet debt service obligations. ▪ To a certain extent it also helps to reduce the exposure of lenders to market risk by improving the DSCR. ▪ In addition, it can also help keep the tariff at affordable levels.
Debt Service Reserve / Cash Deficiency Reserve	▪ Maintenance of debt service in an escrow account (level TBD) can be considered to mitigate part of the revenue volatility / delay in payments. ▪ Promoters provides contractual cash deficiency support commitments for project cash-flow shortfalls.

Conclusion:

- These mechanisms have been tried and tested in different countries with success. It is important for these mechanisms to be flexible enough to adapt to changing market needs (e.g. impact of covid on air traffic) and sector /country characteristics.
- Some of the key considerations financers will focus on are as follows:

➢ Acceptable level of debt service cover from project cashflows: by providing funding viability mechanisms, the quantum of net funds required can be brought down which would improve debt service cover ratio.

➢ Reliability of the counter-party providing the funding viability support: ability and creditworthiness of the counterpart to meet its obligations becomes critical from the lenders' perspective especially if it involves long-term payment commitments (e.g. operation grant, MRG). To address this issue, governments / promoters have typically put in place payment security mechanisms to comfort lenders (e.g. LC, guarantee from 3rd party).

➢ Process to secure funding viability support payments / payment security mechanisms: clear mechanisms to be in place.

Key Challenges & Opportunities

Key Challenges faced in Transport Infrastructure Investment

- Regulatory and Policy Uncertainty
- Project Bankability and Financial Risks
- Land Acquisition and Right-of-Way Issues
- Political and Governance Risks
- Lack of Integrated Regional Planning
- Foreign Investment Restrictions and Local Content Requirements
- Environmental and Social Concerns

To address these challenges, Governments can

- Build institutional capacity for better project preparation and risk management.
- Regulatory Certainty – Strengthen PPP frameworks and provide more risk-sharing mechanisms; Stable policies and clear PPP frameworks attract long-term investors.
- Risk Mitigation – Viability gap funding, guarantees, and concessional financing can improve project bankability.
- Land Acquisition Reforms – Transparent laws and fair compensation models help speed up land acquisition.
- Political Stability – Government commitment and policy continuity ensure project success.
- Regional Cooperation – Enhance regional cooperation for integrated transport networks.Coordinated policies can facilitate cross-border infrastructure development.
- Investment-Friendly Policies – Open foreign investment policies encourage participation from global investors.
- Sustainability Focus – Addressing environmental and social concerns proactively minimizes disruptions.

India’s Push For Infrastructure Development

India emphasizes on improving its transportation network, with various policies which support its development

Toll Roads

Bharatmala Pariyojana

83,677km
Highways and roads to be developed

Rs. 5,35,000cr
Total expected investment

Ports

Sagarmala Project / Maritime India Vision 2030

12
Number of major ports

14
Development of new ports identified

166
Ports identified for modernisation

Airports

5-year plan announced in 2024

50
Planned expansion of airports

5 million
Annual capacity of Bihta Airport

Rail

National Rail Plan Vision

3
Identified Dedicated Freight Corridors by the government

45%
Modal share of railways in freight by 2030

In the latest Union Budget, the Government of India plans to allocate 2.9-3 trillion rupees for railway modernization in 2025/26, expanding tracks and adding 400 high-speed trains. The road network budget will increase by 3-4%, while the Ministry of Road Transport aims to raise 1 trillion rupees annually through tolls and asset monetization.

Source: Economic Times, ENR, Invest India, PIB

Case Studies

05

Case Study: Bulacan Airport Phase 1 PPP

The Bulacan International Airport (the "Project") is a greenfield asset in the coastal area of Bulacan, 35km north of Manila

Philippines

April 2022

US$2.17 billion

Bulacan Airport Phase 1 PPP

Sponsors	▪ San Miguel Corporation
Lender group	▪ 7 banks comprising of SMBC, Standard Chartered, MUFG, KfW, ING, HSBC, Deutsche Bank, backed by Atradius 98%/95% political/commercial cover
Project Size	▪ US$2.1bn
Debt	▪ Term Loan Facility: US$2.17bn
Tenor	▪ 4 years (with an extendable 10 year)
SMBC Net Take	▪ US$300m
SMBC Role(s)	▪ Co-Financial Advisor, Co-Transaction Coordinator, MLA, ECA Agent

Transaction Highlights

- Phase 1 will establish the airport's platform to ensure the area's suitability for airport operations. The project consists of up to 6 runways – with 2 constructed in Phase 1. The airport has an eventual capacity for more than 100 million passengers per annum. The new airport will be 3x larger than the existing Ninoy Aquino international airport in Manila, and is likely to be operational by 2027.
- Phase 2 will include 2 more runways, boosting capacity to 100 million passengers per year.
- San Miguel Aerocity has a 50-year concession under a BOT model. The company in September 2019 received a notice to proceed from the Department ofTransportation after submitting an unsolicited proposal.
- The Project won the Transport Deal of the Year in IJ Global Awards 2022, and involves 7 international lenders and 1 export credit insurer

 Source: IJGlobal

Case Study: Project Walter – Portfolio of Roads in India

Acquisition financing of a portfolio of six operating road assets in India by Actis. The assets are backed by long-term concession arrangements on a Toll or Hybrid Annuity Model (HAM) basis.

The Project

- Acquisition of a portfolio of six roads by Actis, a global PE, from Welspun, a local Indian Infrastructure developer

Underlying assets

- Portfolio of six operating road assets in India comprising 5 assets on Hybrid Annuity concession and one on BOT(Toll) basis
- Hybrid Annuity concession – Availability based pay-outs from state-backed National Highways Authority of India (local AAA credit rating),
- BOT Toll – While the asset exposed to Traffic risk. Tariff risk was managed by **pre-agreed** (without subsequent approvals) tariff escalation formula.

Financing structure

- Total Debt: USD 120m
- Loan repaid over 48 months with a balloon
- First priority security over the issued share capital of the Borrower, inter-company loans between the Parent (as lender) and the Borrower, offshore assets of the Borrower

Key bankability drivers

- Diversified Portfolio: Portfolio comprised six assets, five of which were on hybrid annuity concession i.e., Availability based Payment
- Credible counterparty: Payments are made by National Highways Authority of India (local AAA credit rating)
- Operating Assets with minimal construction Risk: Borrower is insulated from cost overruns, delays, or penalties that may impact toll collection. Seller provided the Borrower indemnity for any loss in revenue due to construction delays.
- Strategic Asset Locations: Assets are located across India, including along the North-South highway corridor. Strong traffic demand is driven by regional / national linkages and by non-discretionary demand from the agriculture and industrial sectors.
- Robust Cashflows: Projected cashflows remain robust even in various stress-tested scenarios, such as discounted market traffic growth. Additionally, Lenders obtain comfort from Leverage and DSCR covenants,

Toll risk managed primarily by a portfolio effect of five other Availability Payment based project

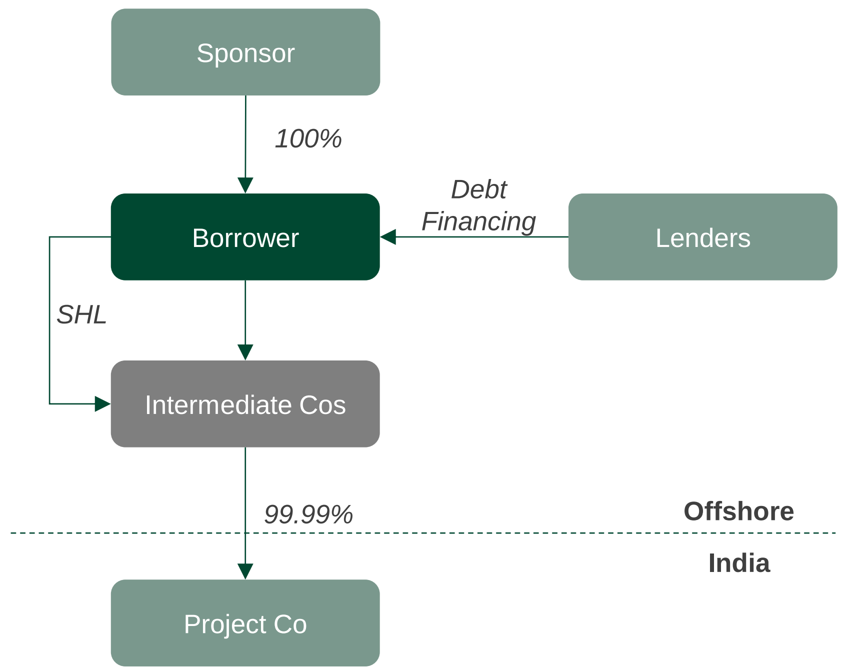
Source: Oliver Wyman, KKR, Statista

Case Study: India Single Toll Asset Acquisition

Financing Connectivity in India – Toll asset connecting the twin cities in India

Project Details	
Project	USD 85 million Holdco Financing for acquisition of an operational toll bridge asset
Buyer	Confidential
Seller	Confidential
Target Acquired	Confidential, with c.99.9% interest in Asset through intermediate Holdcos
Asset	Confidential
Offtake	30-year Concession Agreement with the National Highways Authority of India
Financing Terms	Tenor: 2 years Margin: Confidential Upfront Fee: Confidential
SMBC Roles	Mandated Lead Arranger, Lender and Hedge Provider

Post Acquisition Structure



Case Study – Mexico Line 4 Project

The Guadalajara Línea 4 Project (the "Project") involves the development of the fourth route of an electric train-based mass transport system for one of the largest metropolitan areas in Mexico.

Mexico

June 2024

MXN 5,230.87 MM

Mexico Line 4 Project

Sponsors	▪ Mota-Engil Mexico ▪ CRRC (Hong Kong) Co. Ltd.
Lender group	▪ SMBC, BBVA, and Banobras, the Mexican National Development Bank
Project Size	▪ MXN 9.73BN
Debt	▪ Term Loan Facility: MXN 4,314.22MM ▪ VAT Loan: MXN 916.65MM
Tenor	▪ 15 years
SMBC Net Take	▪ MXN 1,743.62 MM (1/3 of Debt Quantum)
SMBC Role(s)	▪ Structuring bank, Joint Bookrunner, and Hedge Provider

Transaction Highlights

- The Project will connect San Pedro Tlaquepaque with Tlajomulco de Zúñiga municipalities located in the Guadalajara metropolitan area in the Mexican State of Jalisco, and will have a length of ~21.2 km with eight stations.
- The purpose of the Project is to enhance the connectivity of the population of Tlajomulco de Zúñiga and accelerate the economic and commercial development of a historically under-served area.
- The Sponsors and EPC Contractors of the Project are Mota-Engil México, a subsidiary of Portugal's Mota-Engil, in consortium with CRRC (Hong Kong) Co. Ltd, a subsidiary to CRRC.
- This represents the LADA PF Infrastructure and Telecom team's third transaction with Mota-Engil and the second one with CRRC.
- The financing of the Project consists of (i) MXN 4,314.22MM (~USD 253.93MM) 15-year fully amortizing Term Loan, and (ii) MXN 916.65MM (~USD 53.95MM) VAT Loan, for a Debt Quantum Term Loan + VAT Loan) of MXN 5,230.87 MM (~USD 307.88MM)1.
- Similar to prior Project Finance transactions in Mexico, this operation is fully governed by Mexican Law and booked entirely at SMBC SOFOM Mexico.
- SMBC provided pro-rata hedges for the Term Loan including IRS and for the first time an Interest Rate Cap booked at SMBC SOFOM, also fully governed by Mexican Law.

Source: Inframation, PFI,

Case Study: Project Mega

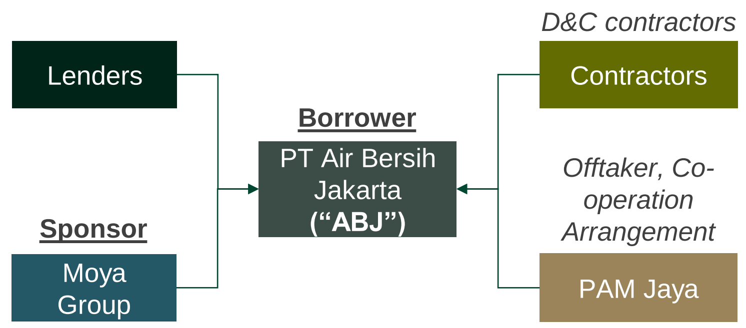
Project Overview

Project	The project includes: ▪ **Brownfield Project:** Maintenance and Rehabilitation of existing 16,800 lps water treatment plants, ▪ **Greenfield Project:** 1. Construction of a new water treatment plant with 3,000 lps capacity and 2. Installation of new pipe networks and new connections to connect the unserved areas in Jakarta to existing pipe network The project aims to increase Jakarta's tap water coverage to 100%, ensuring all residents have access to clean piped water by 2030
Sponsor	▪ PT Air Bersih Jakarta, a company of Moya Asia group
Offtake	▪ Offtake: 25 years Co-operation Agreement with local municipal body PAM Jaya, responsible for supplying water to entire Jakarta.
Timeline	▪ Facility signing in February 2023. ▪ Construction under the Greenfield Project is expected to be implemented in stages over a period of 5 years
Project Cost	▪ Total capital expenditure of approximately c.IDR 26.7 Tn

Project Overview (Continued)

Financing	▪ Loan: IDR 8.87 Tn in total
Lending Group	▪ Bank Tabungan Pensiunan Nasional (BTPN) ▪ Bank Central Asia ▪ OCBC Bank ▪ Bank OCBC NISP ▪ Bank Tabungan Negara (Persero) ▪ KB Bukopin ▪ China Construction Bank Indonesia ▪ Sarana Multi Infrastruktur (Persero)

Structure Chart



Source: PAM Jaya, BTPN website

SMBC Credentials

06

Coverage Across the Infrastructure Sector

- SMBC Group has extensive experience of financial advisory, lead arranging, underwriting & bond lead managing on infrastructure projects globally.
- We are considered an experienced funding institution in the development of infrastructure finance. SMBC provides extensive coverage across the Transportation, and other public-private partnership infrastructures

Airport Infrastructure

Development of airside (e.g. runways, cargo, hangars), landside (e.g. parking lots, power supply) and airport support facilities.

Toll Road Infrastructure

Development of toll roads including barrier toll collection, entry/exit toll collection and electronic toll collection (e.g. dedicated short range communications, automatic number plate recognition)

Ports Infrastructure

Development of ports (including inland port, fishing port, dry port, warm water port and seaports)

PPP Infrastructure

Development of public infrastructure via a public-private partnership, including stadiums / sports hubs, water treatment PPP, waste-to-energy PPP, incineration plants, etc.

Overview of the Transportation Infrastructure Sector

Airport Infrastructure Industry Trends

- Liberalization of air transport services, as a result of (i) changing traffic patterns due to changing flight routes and (ii) increase in passenger traffic, is driving expansion of facilities and renovation of secondary airports to absorb growth
- Asia-Pacific to witness the highest growth rate in the next five years as airport infrastructure builds have been slow despite growing air passenger traffic.
- Commercial airport segment is expected to dominate the market as this segment is expected to see the fastest growth in APAC
- China is expected to lead the APAC market in terms of growth as the China Aviation Administration of China plans to construct 200 new airports to reach its target of 450 airports

Toll Road Infrastructure Industry Trends

- Government support for road concessions (e.g. subsidizing cost of land, revenue support, etc.) are driving growth in the toll road infrastructure sector in places like India.
- Multilateral agencies like the Asian Development Bank (ADB), Asian Infrastructure Investment Bank (AIIB) and World Bank are playing an increasingly important role in catalysing sector reforms and helping develop the pipe of economically viable and environmentally sustainable projects, especially for countries with financially strained governments.
- With the Covid-19 pandemic, personal transport has become the preferred mode of transport for health and safety reasons. As such, toll-road traffic has fully recovered by 2022-23, which is much faster than air traffic’s expected full recovery by 2024-25.

Port Infrastructure Industry Trends

- Government initiatives for port development are driving growth – In India, the government has envisioned a total of 189 projects for modernisation of ports involving an investment of Rs 1.42 trillion (USD 22 billion) by the year 2035 under the Sagarmala Programme.
- Asia Pacific is expected to be the fastest growing markets for the port infrastructure industry, especially for China and India as an increasingly larger portion of international trade takes place via marine transport, driving the demand for more port infrastructure.
- Ports are increasingly implementing innovations such as Robotics, big data analytics and predictive maintenance across the entire value chain (from approach services to loading/unloading to storage to industrial processing).
- Efforts to reduce the carbon footprint and reduce environmental impact of the maritime sector remain high on the international agenda.

 Sources: Deloitte Global Port Trends 2030, ICAO, Mordor Intelligence, PPIAF

Coverage Across the Infrastructure Sector

India 2024
Actis Acquisition of Bridge Road Assets
USD 85M
MLA & Bookrunner

India 2022
Actis Acquisition of Road Asset Portfolio
USD 120M
MLA

Hong Kong 2018
Hong Kong Route 3 Refi
HKD 1.54B
MLA

Vietnam 2013
National Highway 20 Project
USD 250M
MLA

SEA Ongoing
SEA International Airport
Ongoing
FA

Myanmar 2015
Airport PPP Project
FA

Vietnam 2012
Long Thanh International Airport
FA

Sri Lanka 2016
Colombo Container Port Terminal
FA

Vietnam 2012
Cai Mep Container Terminal
FA

Sources: Deloitte Global Port Trends 2030, ICAO, Mordor Intelligence, PPIAF

Selected Asia-Pacific PPP Infrastructure Credentials

PPP Infrastructure

: Sports | : Power | : Water | : Incineration

Indonesia 2023
Jakarta Drinking Water Project
IDR 8,874B
MLA (BTPN)

Taiwan 2023
Framosa Solvent Recovery Plant
NTD 1,350B
MLA, Green Loan Coordinator

Vietnam 2019
Municipal Water Treatment PPP
FA

Singapore 2014
Singapore NEWater Project
FA

Malaysia 2014
Waste-to-Energy PPP
FA

Singapore 2013
Senoko Incineration Plant
FA

India 2011
Dahej Desalination Plant
FA

Singapore 2007
Singapore Sportshub
SGD 150M
FA, MLA

Source: Inframation, PFI

Selected Global Transportation Infrastructure Credentials

Transportation Infrastructure

: Transportation

Country	Year	Deal	Amount	Role
Mexico	2024	Mexico Line 4	MXN 5,230.87MM	Structuring bank, Joint Bookrunner, Hedge Provider
UK	2024	HS1 LCF	GBP 48.8MM	MLA, Hedge Provider
Mexico	2022	Metro Line 1	MXN 12.97BN	MLA, Hedge Provider
Spain	2021	Barcelona L9 Section 1	EUR 1,174MM	MLA
Australia	2021	QNGR PPP	A$638MM	MLA
UK	2021	Angel Train Refinancing	GBP 600MM	Lender
Australia	2020	Canberra LRT Metro PPP	A$286MM	MLA
Spain	2020	Barcelona L9 Section 4	EUR 720 million	Lender
France	2020	Synerail Refinancing	EUR 54 MM	MLA
France	2019	Ermewa Refinancing	EUR 1,100 million	Lender
UK	2019	Beacon Rail Refinancing	EUR 1,200 million	Lender, Initial Arranger
Ireland	2019	DISA Asset Acquisition	EUR 194 million	MLA, Bookrunner, Sole Placement Agent
UK	2019	Elizabeth Line (Class 345)	GBP 1,100 million	MLA, FA
UK	2019	Wales and Borders Rolling Stock	EUR 392 million	MLA, Hedging Bank
France	2019	CDG Express Rail	EUR 620 million	Bookrunner, MLA
Australia	2018	Melbourne Metro Tunnel PPP	A$3,350MM	LA
Australia	2015	Sydney Light Rail PPP	A$1,550MM	LA
Australia	2006	Reliance Rail PPP	A$3.6BN	MLA

 Source: Inframation, PFI

Thank You

Breakout Session:
Brainstorming and deep dive on the projects presented, including feedback and suggestions on improving the projects

Breakout Session

	Group A	Group B	Group C	Group D	Group E
Country	**Lao PDR**	**Cambodia**	**Philippines**	**Malaysia**	**Vietnam**
	Indonesia	**Singapore**	**Brunei**	**Myanmar**	**Thailand**
	Timor Leste	n/a	n/a	n/a	n/a
Facilitator	**Mr. Visheh Gupta**	**Ms. May (Tint Tel May)**	**Mr. Andy Sutrisno**	**Mr. Junxian Shi**	**Mr. Julien Venner**
Meeting Room	**Plenary - Right**	**Plenary – Left**	**Penang Room**	**Kelantan Room**	**Perak Room**

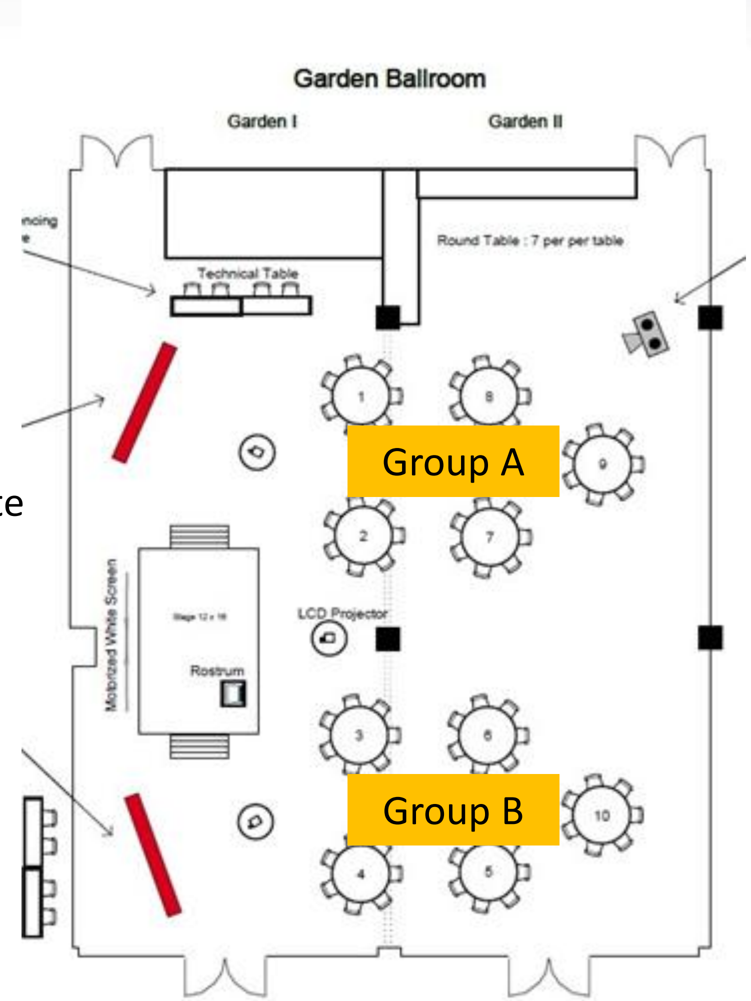
Instructions:

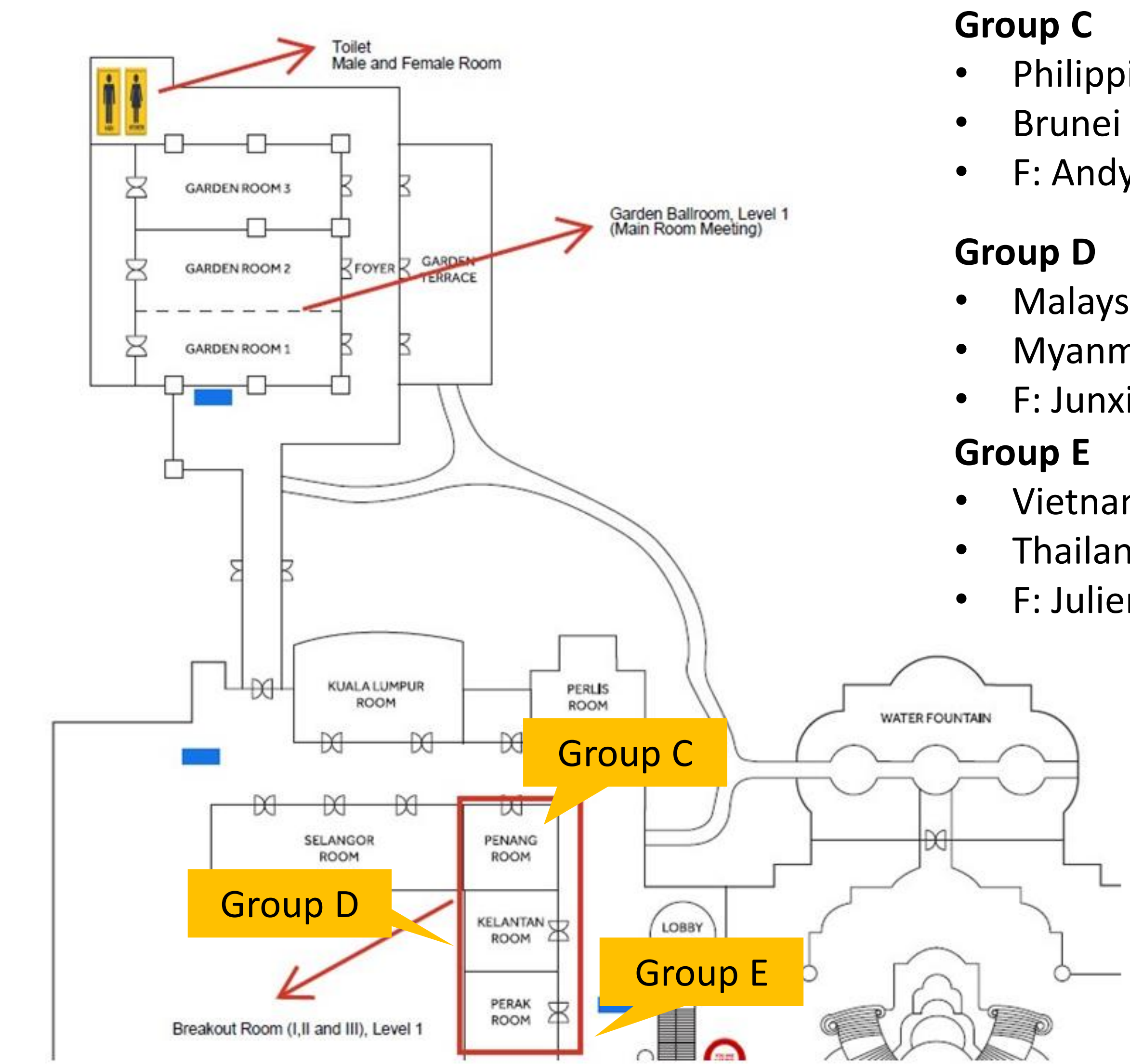
- *Please select a **group leader** and reflect on each of the potential projects presented and discuss as per instructions in the "Guiding Instructions From Facilitators" slide.*
- *Please make note of the conclusion your team makes for each questions/topics.*
- *Participants may also wish to share relevant experience on past or ongoing similar projects in their country, if any.*
- *Kindly proceed to your assigned meeting room when ready.*

Breakout Session – Layout

Group A
- Lao PDR
- Indonesia
- Timor Leste
- F: Visheh

Group B
- Cambodia
- Singapore
- F: May





Group C
- Philippines
- Brunei
- F: Andy

Group D
- Malaysia
- Myanmar
- F: Junxian

Group E
- Vietnam
- Thailand
- F: Julien

Guiding Instructions from Facilitators

Project-related Discussions

Based on the project presentations, how does each presented projects align with the assigned criteria?
Please spend 20 mins on discussions and 10 mins on summarizing the findings on the flipboard provided for each teams.

Team	Assigned Criteria
Team A	A) strategic relevance, B) regional connectivity, and C) resilience
Team B	A) regional connectivity, B) resilience, C) economic, environmental & social benefits
Team C	A) resilience, B) economic, environmental & social benefits, and C) project feasibility
Team D	A) economic, environmental & social benefits, B) project feasibility, and C) implementation capacity
Team E	A) strategic relevance, B) project feasibility, and C) implementation capacity

General Discussions

Please discuss the following. Please spend around 20 mins on discussions and 10 mins on summarizing the findings–

- What risks do you foresee in finalizing your proposals?
- How do you intend to address these risks?
- In what way can the Project (implemented by SMBC) support you in addressing some of these risks?

Please report back to Plenary at 15:15.

Breakout Session Takeaways

	Group A	Group B	Group C	Group D	Group E
Country	**Lao PDR**	**Cambodia**	**Philippines**	**Malaysia**	**Vietnam**
	Indonesia	**Singapore**	**Brunei**	**Myanmar**	**Thailand**
	Timor Leste	n/a	n/a	n/a	n/a
Facilitator	Mr. Visheh Gupta	Ms. May	Mr. Andy Sutrisno	Mr. Junxian Shi	Mr. Julien Venner

Instructions:
Each group leader is invited to present the findings and conclusion from the group.

Coffee Break
(15 mins)

Informational session:
key developments, challenges, and opportunities in digital infrastructure in the region

Disclaimer

a) Information and opinions in this document should not be construed as legal, regulatory, financial, investment, tax, or accounting or other technical or professional advice.

b) Recipients should make their own independent appraisal and/or risk assessment of the information and opinions and consult their professional advisors before embarking on any course of action.

c) Nothing contained in this document is intended to be, nor should be construed as, a recommendation, solicitation or an offer of any financial products, investments, financing or other banking facilities.

d) Although the information is obtained from sources considered as reliable by Sumitomo Mitsui Banking Corporation ("SMBC"), SMBC makes no representation or warranty as to, and accepts no liability for any representation in relation to, the accuracy or completeness of the information and opinions contained in this document or for any loss or damage arising from any reliance on or use of this document. SMBC has no responsibility to update any information or opinions contained in this document.

e) The contents of this document are confidential and shall not be disclosed to any other party without SMBC's prior written consent. The contents contained in this document may change without prior notice.

Note: Certain images may be subject to copyright. Maps are for illustrative purposes only.

Sumitomo Mitsui Banking Corporation Singapore Branch, incorporated in Japan with limited liability (Reg. No (UEN) T03FC6366F)

01

Overview of Digital Infrastructure in ASEAN

Disclaimer

This document has been prepared by Sumitomo Mitsui Banking Corporation (“SMBC”) for the sole purpose of information sharing.

Information and opinions in this document should not be construed as legal, regulatory, financial, investment, tax, or accounting or other technical or professional advice. Recipients should make their own independent appraisal and/or risk assessment of the information and opinions, and consult their professional advisors before embarking on any course of action.

Nothing contained in this document is intended to be, nor should be construed as, a recommendation, solicitation or an offer of any financial products, investments, financing or other banking facilities.

Although the information is obtained from sources considered as reliable by SMBC, SMBC makes no representation or warranty as to, and accepts no liability for any representation in relation to, the accuracy or completeness of the information and opinions contained in this document or for any loss or damage arising from any reliance on or use of this document. SMBC has no responsibility to update any information or opinions contained in this document.

The contents of this document are confidential and shall not be disclosed to any other party without SMBC’s prior written consent. The contents contained in this document may change without prior notice.

Where third party products or services are referred to, they are for information/illustration purposes only and are not a recommendation or endorsement by SMBC. Recipients should make their own independent appraisal of and decision to work with any third party or on the use of their products or services.

SMBC Nikko Securities (Hong Kong) Limited, SMBC Nikko Capital Markets Limited, SMBC Nikko Securities Inc., SMBC Nikko Securities and their affiliates (“SMBC Nikko”) are related companies of SMBC. Information and presentations by SMBC Nikko are made in their own right and SMBC is not authorized to, and does not hold itself out as, representing SMBC Nikko or promoting or entering into products and services for or on behalf of SMBC Nikko. SMBC has not verified the information or materials provided by SMBC Nikko and accepts no responsibility in relation thereto.

Note: Certain images may be subject to copyright. Maps are for illustrative purposes only.
Sumitomo Mitsui Banking Corporation Singapore Branch, incorporated in Japan with limited liability (Reg. No (UEN) T03FC6366F)

Data Demand and Drivers

ASEAN is experiencing a rapid spike in data demand (20% CAGR over the next 5 years) driven by increased digital adoption/transformation

ASEAN is experiencing digital economy expansion and transformation...

Demand Drivers

Increasing Digitalization in Southeast Asia

- Rising digitalization in the region, evident from **SEA's surging digital economy (e.g. GMV doubled from USD102bn to USD219bn between 2019 and 2023)**, has led to exponential increase in data volume / consumption, necessitating the rapid expansion of infrastructure to meet data storage, processing and management needs

Higher Enterprise Spending on Cloud Services & Advanced Technologies

- Cloud services and advanced technologies (e.g. AI) offers enterprises benefits such as **reduction in costs, scalability / flexibility and greater collaboration / productivity.** Accordingly, enterprises are increasingly digitizing operations

Advanced Data Analytic Use Cases

- The rising adoption of advanced data analytics has drove demand for computation capacity to process, analyze, and store large datasets in real time, fueling the growth of the global data analytics market **from USD 59bn in 2019 to USD 95bn in 2023.**

Supply Drivers

Infrastructure Upgrades

- Rising demand has fueled SEA infrastructure development, leading to outsized SEA bandwidth growth (contributing to **~40% of cross-Asian international bandwidth expansion)**. This was driven by investments in subsea cables and power assets, enabling the construction of more globally interconnected data centers.

Strong Push from Hyperscalers & Colocation Providers

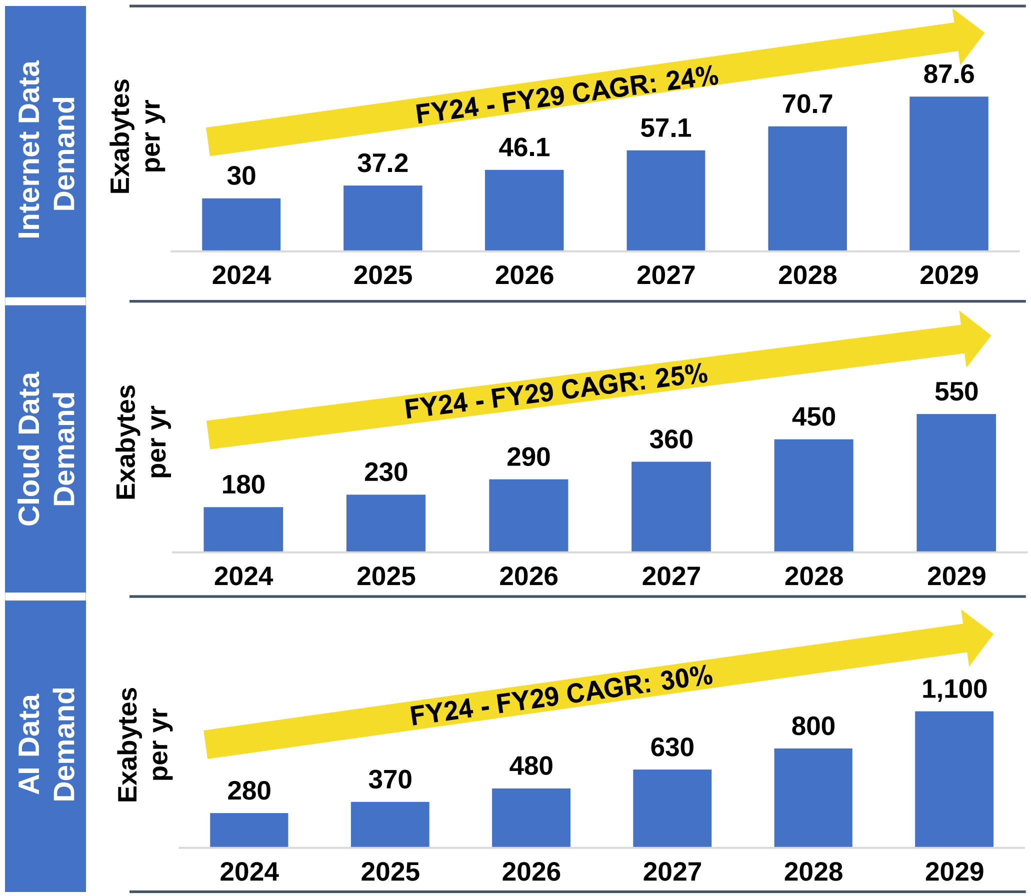
- SEA's emergence as a digital hub has attracted **strong investment from hyperscalers and colocation providers**, driving scalability, efficiency, and best-in-class capabilities in the region.

Data Protection Regulations & Policies

- Governments in SEA have implemented moderate data protection regulations, which has led to cloud and infrastructure providers **establishing large digital infrastructure footprints domestically**

...contributing to burgeoning data traffic and demand in SEA

Market Demand and Growth (2024 – 2029)

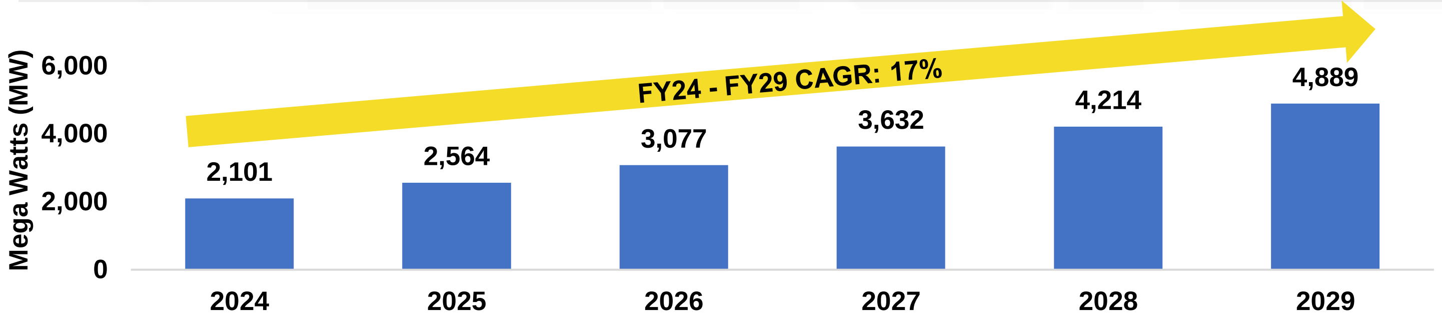


Sources: Arizton, BCG, Ericsson Mobility Report, GSMA, World Broadband Association, asean.org, Telecom Review Asia, Statista, ASEAN Digital Masterplan 2025, Gartner

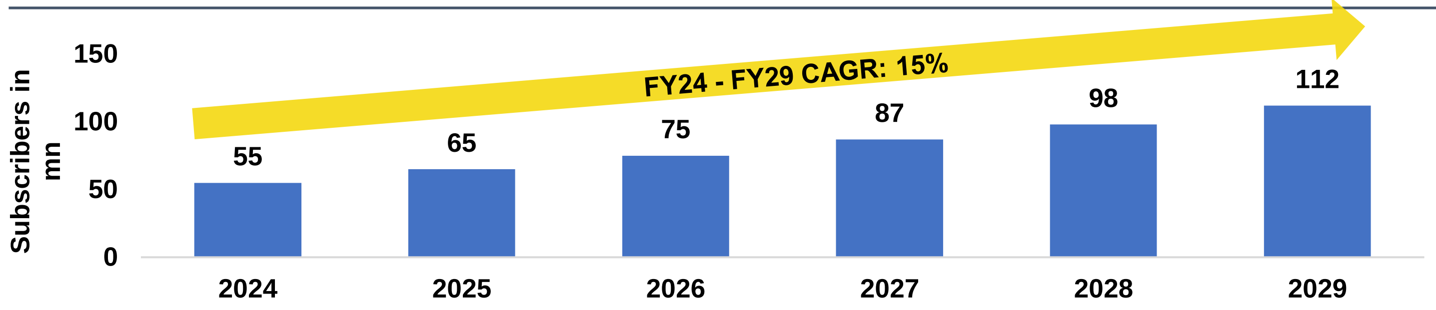
Spiking Digital Infra Demand – DC / Fiber / Towers

Market Demand (2024 – 2029)

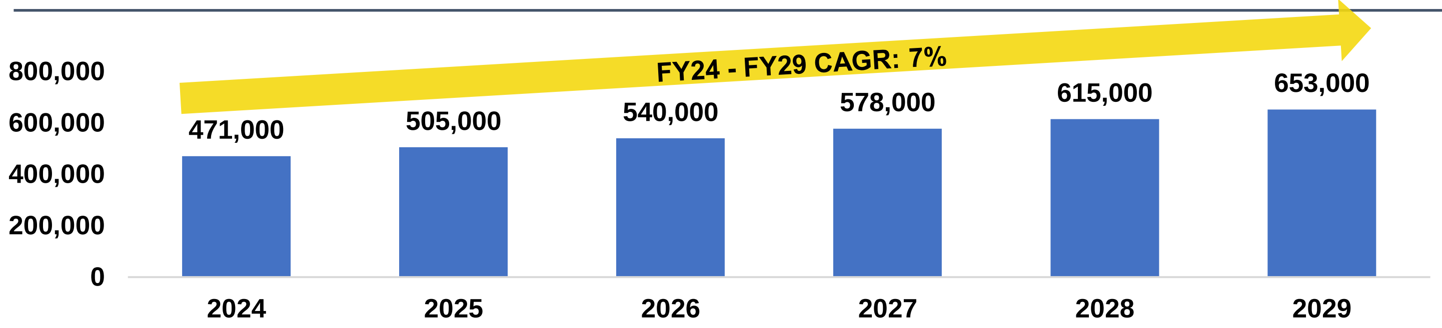
Data Centres



Fibre (FTTH)



Telecom Towers



Commentary

- DC market demand is expected to grow at **17% CAGR** from 2024 to 2029
- Demand mainly driven by increasing cloud adoption, AI workloads and OTT stream & content delivery
- Growth in SEA markets has accelerated post-2019, with Malaysia and Indonesia in particular experiencing increased investment from cloud/AI players (and to a certain extent spillover demand from Singapore), while Thailand and the Philippines have untapped data centre development potential

- Fibre market demand, particularly Fibre To The Home (FTTH), is expected to grow at **15% CAGR** from 2024 to 2029
- This has been spearheaded by governments which are seeking to boost broadband access to underserved rural and suburban areas. 5G backhaul requirements has also been a key driver, especially for dense small cell 5G deployments

- Telecom Towers market demand is expected to grow at a **7% CAGR** from 2024 to 2029
- This is expected to be driven by :
  - 5G network rollouts which requires higher tower density than 4G
  - Rural connectivity initiatives such as Indonesia's BAKTI and Malaysia's JENDELA,
  - Increasing fiber backhaul being deployed alongside towers

Sources: DC Byte, Cushman Wakefield Global DC Market Comparison, Knight Frank DC Report SEA Edition 2023, Towerxchange, GSMA Intelligence & Mobile Economy Reports, FTTH Council APAC Report, asean.org, adb.org, MIC Vietnam, KOMINFO, DICT Philippines, ONDE Thailand, IMDA, MCMC

Potential Obstacles to Developing Digital Infra in SEA

Challenges	Description
Regulatory Compliance	▪ Regulatory frameworks across ASEAN are outdated or inconsistent, making it difficult for telco and cloud providers to keep up with evolving technology and deployment needs ▪ Cumbersome permit processes, foreign ownership limits, and varying national rules delay infrastructure rollout ▪ Lack of regional harmonisation, such as differing spectrum standards and data localization laws creates cross-border inefficiencies and complicates cloud expansion
Investment and Financing Constraints	▪ SEA telcos require an estimated US$40-60bn in additional investment over the next five years to expand fiber, 5G and data centre infrastructure but low ARPUs, high interest rates and weak local currencies render financing such investments challenging ▪ While some progress has been made with independent TowerCos and DC operators that facilitate infrastructure sharing, less developed markets still face funding challenges, risking delays in critical infrastructure rollouts like FTTH and 5G
Connectivity Gaps and Geographic Barriers	▪ SEA digital connectivity remains uneven, with rural areas in Cambodia, Indonesia and Myanmar lagging far behind urban centres in broadband/mobile access ▪ Geographic factors such as archipelagos, uneven terrain and limited electricity infrastructure make last-mile fibre and tower deployment expensive and logistically challenging
Talent Shortages and Skills Gap	▪ There is currently a lack of skilled talent in critical areas like cloud architecture and network engineering. This has slowed infrastructure deployment, impacting service quality and innovation ▪ While upskilling initiatives and cross-border talent mobility have alleviated this somewhat, intense global competition for tech talent means SEA must urgently invest in education and training to meet its digital infrastructure goals
Environmental Constraints	▪ Growing power requirements (especially for data centres) pose power reliability and sustainability issues ▪ Countries like Singapore have imposed restrictions on new data centres due to land and power limitation, highlighting regional struggles with grid capacity and limited renewable energy
Cybersecurity and Infrastructure Resilience Risk	▪ Rapid digital infrastructure expansion in SEA increases exposure to cyberattacks, data breaches and physical disruptions with many countries still developing robust cybersecurity frameworks ▪ Key vulnerabilities include limited network redundancy, submarine cable risks, inconsistent cyber defence capabilities and unprotected growth in AI/IoT ecosystems
Cross Border Integration and Geopolitical Factors	▪ The absence of a unified digital single market increases operational cost for cloud platforms and reduces regional efficiency ▪ Geopolitical tension and varying alliances (e.g. US-China tech rivalry) lead to incompatible infrastructure choices and create diplomatic and technical barriers to regional network integration

Sources: GSMA – Mobile Policy Handbook & Spectrum Reports & Mobile Connectivity Index, ASEAN Digital Masterplan 2025, World Bank – Digital Economy for ASEAN Report & Securing the Digital Economy in ASEAN, unescap.org, GSMA –Mobile Connectivity Index, Interpol.int

Digital Infra Financing Considerations

1 Debt Sizing
- Projects with higher profitability / stable cashflows preferred, especially with highly creditworthy tenants (e.g. US hyperscalers)
- Robust offtake termination for convenience mitigation is a key consideration for lenders

2 Financial covenants
- Lenders will seek to optimize debt size while allowing for an adequate buffer in the event of financial deterioration
- Debt sizing approaches are typically based on: 1) project cashflows; 2) loan-to-value / loan-to-cost; and 3) leverage ratios

3 Underlying operational considerations
- Financial covenants provide an avenue to monitor the borrower's financial performance and allow lenders to intervene in an adverse scenario
- Typical financial covenants, include but are not limited to: 1) Debt service cover ratio; 2) Interest cover ratio; 3) Net leverage ratio; and 4) Loan to value

4 Loan tenor and repayment
- Typical loan tenors range from 3 – 7 years
- Repayment typically structured with a portion of total debt amount is amortised with a sizeable bullet at maturity to accommodate construction period
- In certain transactions, cash sweep mechanism whereby excess cash flow is used to prepay the debt can be considered

5 Portfolio maturity
- Portfolio financings are generally considered less risky, especially when there are stabilized assets within the portfolio i.e. less construction risk and more cash generative
- Generally, portfolio financings in the DC sector are starting to emerge as opposed to single asset financing

6 Security
- Typically includes all material assets (e.g. key contracts, insurances and account balances) and all shares (and shareholder loans) of the borrower/operating company

7 Guarantees
- Guarantees might be granted by the sponsor to cover project cost overruns or interest shortfall
- Typically negotiated subject to perceived construction risk, market risk and/or weak contractual protection from the contractors as reviewed by the lenders' advisers

Key lender considerations
Digital Infra debt transactions

Legend :
- Financial metrics / considerations
- Loan structuring considerations
- Security / guarantee considerations

02

SMBC Credentials

Digital Infrastructure Coverage in APAC

SMBC Aus4ASEAN FUTURES asean

Dedicated specialist team

- Core team of experienced professionals with deep sector expertise and robust structuring experience in the sector to advise on your financing requirements

Highly Active in Digital Infra Financing

- SMBC has numerous track record of data center financings globally
- In APAC, SMBC is the most active DI lender in recent years, having closed 21 deals as the MLA / Lender since 2021.

One-stop shop for your banking needs across APAC

- Strong capability to underwrite / coordinate debt financing in both local and international currencies
- Able to provide a complementary suite of banking services in addition to financing (e.g. hedging solutions, transaction banking, agency roles)

Selected Recent Digital Infrastructure Credentials in APAC

Malaysia	Australia	India	Singapore	India	Indonesia
Refinancing & capex financing of a DC portfolio	**AirTrunk DC portfolio acquisition financing**	**Telco tower acquisition financing**	**Princeton Digital Group refinancing & upsizing of DC**	**AdaniConnex greenfield DC portfolio**	**Fiber portfolio acquisition financing**
c.USD 3bn	**c.AUD 24bn[1]**	**c.USD 940mn**	**c.USD 100mn**	**c.USD 880mn**	**c.IDR 1.5tn**
MLA	*Financial Advisor, Lead Structuring Bank, MLAB*	*MLA*	*MLAB*	*MLA, SLL Co-Coordinator, Agent, Account Bank*	*MLA*
2025	2024	2024	2024	2024	2023

Indonesia	Indonesia	Southeast Asia	India	Singapore	Indonesia
Acquisition of PT Indointernet Tbk	**ECX's greenfield DC project**	**Acquisition & capex financing of a DC portfolio**	**AdaniConneX greenfield DC portfolio**	**Greenfield hyperscale DC for a global DC operator**	**Multi-phased greenfield DC project**
c.USD 110mn	**c.USD 310mn**	**Confidential**	**c.USD 210mn**	**c.SGD 530mn**	**c.USD 170mn**
MLA, Sole Hedge Bank	*MLA*	*Lender*	*MLAB*	*MLA*	*Sole MLAB*
2023	2023	2023	2023	2022	2021

Source: PFI, IJGlobal, Inframation, Loan Connector, Company Websites
Note: (1) Represents 100% of the enterprise value for AirTrunk, rather than the financing quantum.

Sustainable Energy, Infrastructure & Export Finance

*SMBC is an award winning and **leading Sustainable Energy & Infrastructure Bank in Asia** supporting our clients requirements across core sectors with multiple product offerings*

Awards & Credentials

2023 Asia-Pacific Bank of the Year
2024 Global Bank of the Year (*eight times in total)*
PFI | 2024, 2021, 2020, 2019, 2017, 2014, 2012, 2008

PFI AWARDS 2024

Project Finance House of the Year
Global, APAC, Hong Kong, Japan & Taiwan
The Asset Triple A | 2024

Selected Credentials

2023: APAC Deals of the Year
Wind: Monsoon Wind
Offshore Wind: Hai Long
Petchem: Balikpapan
Acquisition: CWP Renewables
Digital: AdaniConneX

PFI AWARDS 2023

Global Deals of the Year
Green Deal – Neom Helios
ESG Deal – AMEA Power
M&A Deal – Hornsea 2

Key Strengths:

Sector specialist teams

- Core team of professionals in APAC, covering core sectors: Power, Renewables, New Energies such as Hydrogen, Biofuels, CCUS & EV Mobility, Metals & Mining, Infrastructure, TMT and Healthcare
- Over 300 professionals globally to provide a wide range of services in structured finance

Significant debt underwriting and DCM capability[1]

- Large and diverse Sustainable Infrastructure and Energy asset book
- Deep knowledge base and appetite to continue lending to Sustainable Infrastructure and Energy deals including project bonds in collaboration with SMBC Nikko[1]

Leading advisor on projects throughout APAC

- Proven experience in structuring complex transactions, in-depth knowledge of bankable contractual and financing structures, and strong understanding of lenders' requirements.

Sources: Refinitiv, PFI, The Asset, IJGlobal
Note: (1) Capital Markets activities are carried out exclusively by SMBC's capital markets affiliate, SMBC Nikko

Q&A

This is the end of Day 1 workshop.
Thank you for your active participation.
See you tomorrow!

Regional Workshop on Updating and Advancing the Initial Pipeline of ASEAN Infrastructure Projects

9 – 10 April 2025

10th April 2025

Workshop – Day 2

Agenda

Time (GMT+8)	Agenda	Format
0830 – 0900	Registration for Day 2	
0900 – 0915	Opening by SMBC to summarise Day 1 discussions	Plenary
0915 – 1025	Presentations by project owners on potential projects (3-4 projects) (Note: 15min presentation + 5min Q&A for each Project)	Plenary
1025 – 1035	Coffee break	
1035 – 1135	Brainstorming and deep dive on the projects presented, including feedback and suggestions on improving the projects	Breakout
1135 – 1205	Discussion of feedbacks and suggestions from Breakout session	Plenary
1205 – 1215	Wrapping up of Day 2 and next steps	Plenary
1215 – 1225	Event Online Survey	Plenary
1225 – 1230	Closing remarks by The ASEAN Secretariat	Plenary
1230 – 1400	Lunch	
1400 – 1530	Further follow up / bilateral meetings between interested ASEAN Member States and project consultant SMBC to further discuss and provide further support on project submissions (optional)	

Reflection from Day 1

Facilitated by
Mr. Satya Ramamurthy
(Chair of Public Sector, SMBC)

Presentations by project owners on potential projects

Landbridge Project
(Thailand)

North Mindanao High Standard Highway (Philippines)

SEARECC National Road No.2 Upgrading Project (2W&2E)
&
Upgrading National Road No.8
1 as part of the Asian Highway (AH 15) Network

(Lao PDR)

Coffee Break
(15 mins)

Breakout Session

	Group A	Group B	Group C	Group D	Group E
Country	**Lao PDR**	**Cambodia**	**Philippines**	**Malaysia**	**Vietnam**
	Indonesia	**Singapore**	**Brunei**	**Myanmar**	**Thailand**
	Timor Leste	n/a	n/a	n/a	n/a
Facilitator	**Mr. Visheh Gupta**	**Ms. May (Tint Tel May)**	**Mr. Andy Sutrisno**	**Mr. Junxian Shi**	**Mr. Julien Venner**
Meeting Room	**Plenary - Right**	**Plenary – Left**	**Penang Room**	**Kelantan Room**	**Perak Room**

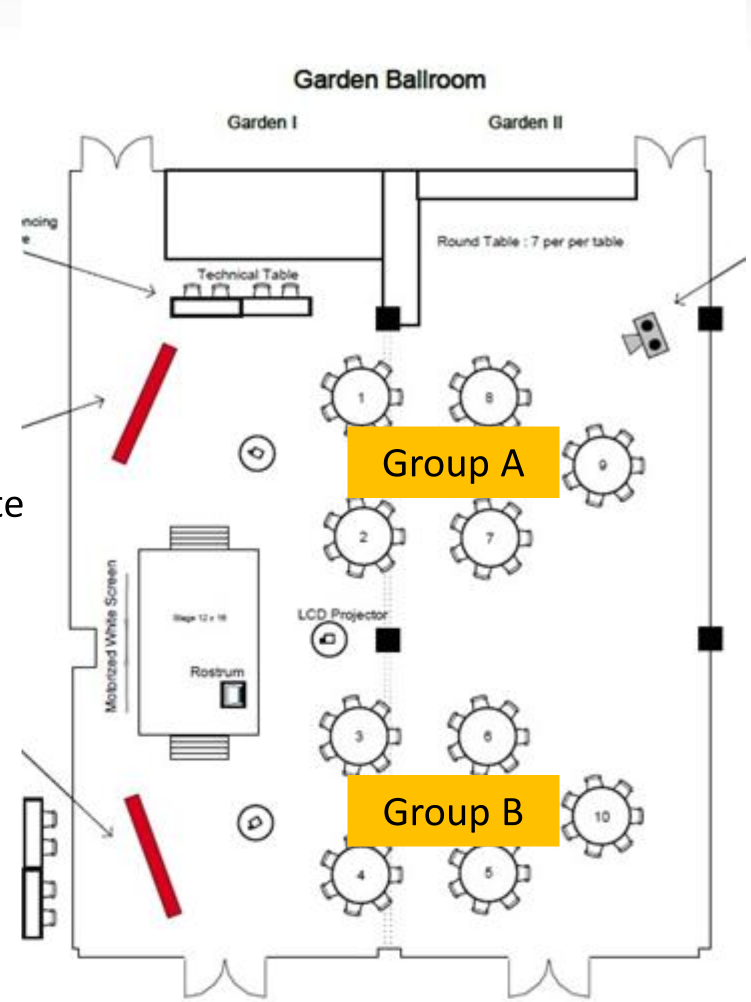
Instructions:

- *Please select a **group leader** and reflect on each of the potential projects presented and discuss as per instructions in the "Guiding Instructions From Facilitators" slide.*
- *Please make note of the conclusion your team makes for each questions/topics.*
- *Participants may also wish to share relevant experience on past or ongoing similar projects in their country, if any.*
- *Kindly proceed to your assigned meeting room when ready.*

Breakout Session – Layout

Group A
- Lao PDR
- Indonesia
- Timor Leste
- F: Visheh

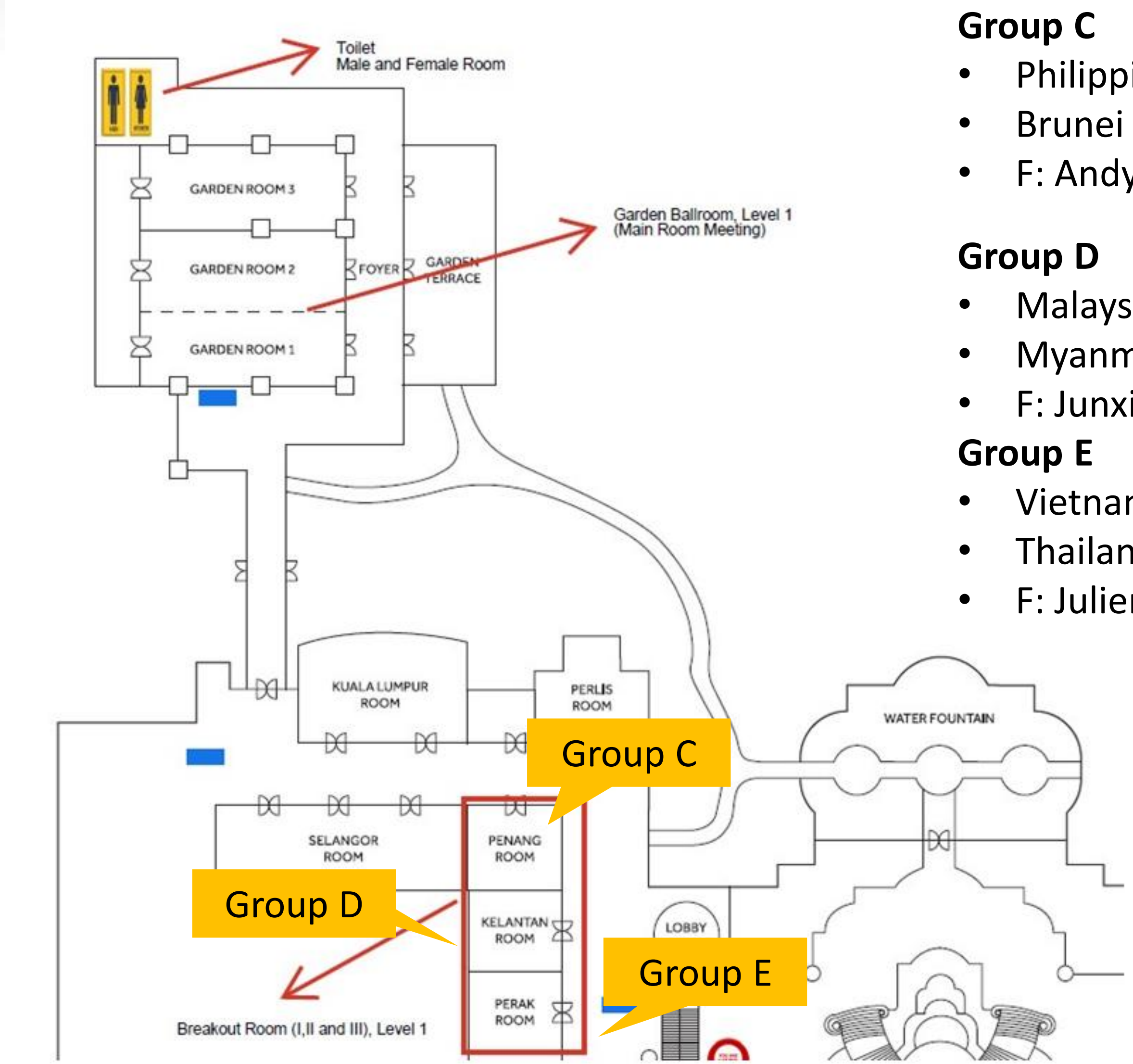
Group B
- Cambodia
- Singapore
- F: May



Group C
- Philippines
- Brunei
- F: Andy

Group D
- Malaysia
- Myanmar
- F: Junxian

Group E
- Vietnam
- Thailand
- F: Julien



Guiding Instructions from Facilitators

Project-related Discussions

Based on the project presentations, how does each presented projects align with the assigned criteria?
Please spend 20 mins on discussions and 10 mins on summarizing the findings on the flipboard provided for each teams.

Team	Assigned Criteria
Team A	A) strategic relevance, B) project feasibility, and C) implementation capacity
Team B	A) economic, environmental & social benefits, B) project feasibility, and C) implementation capacity
Team C	A) regional connectivity, B) resilience, C) economic, environmental & social benefits
Team D	A) resilience, B) economic, environmental & social benefits, and C) project feasibility
Team E	A) strategic relevance, B) regional connectivity, and C) resilience

General Discussions

Please discuss the following. Please spend around 20 mins on discussions and 10 mins on summarizing the findings.

- What risks do you foresee in finalising your proposals?
- How do you intend to address these risks?
- In what way can the Project (implemented by SMBC) support you in addressing some of these risks?

Please report back to Plenary at 11:35.

Breakout Session Takeaways

	Group A	Group B	Group C	Group D	Group E
Country	**Lao PDR**	**Cambodia**	**Philippines**	**Malaysia**	**Vietnam**
	Indonesia	**Singapore**	**Brunei**	**Myanmar**	**Thailand**
	Timor Leste	n/a	n/a	n/a	n/a
Facilitator	Mr. Visheh Gupta	Ms. May	Mr. Andy Sutrisno	Mr. Junxian Shi	Mr. Julien Venner

Instructions:
Each group leader is invited to present the findings and conclusion from the group.

Closing Remarks and Next Steps by

Mr. Satya Ramamurthy
Chair of Public Sector, SMBC

Closing Remarks by

Mr. Lim Chze Cheen

Director, ASEAN Connectivity Division,
ASEAN Secretariat

Online Survey

Instruction: Please kindly scan the QR code and fill the survey on this Regional Workshop

This is the end of the ASEAN Regional Workshop.
Thank you for your active participation.
Please kindly proceed for lunch!

Note: After lunch, there will be further follow up / bilateral meetings between interested ASEAN Member States and project consultant, SMBC, to further discuss and provide further support on project submissions.
Please see your facilitator after lunch.
(optional)